SFU NatLangLab

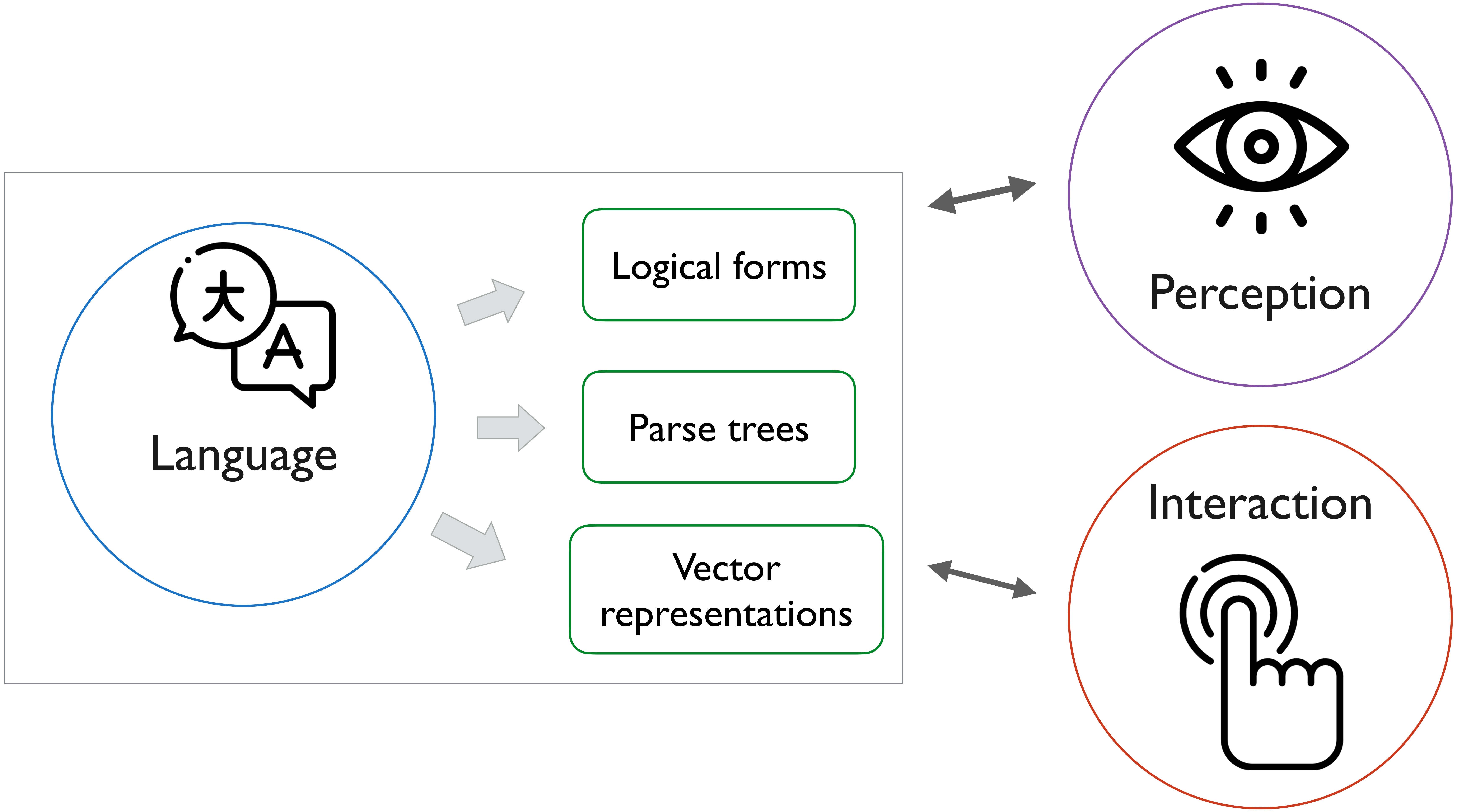
CMPT 713: Natural Language Processing

Grounded Natural Language

Spring 2023

2023-04-05

Semantics does not exist in isolation

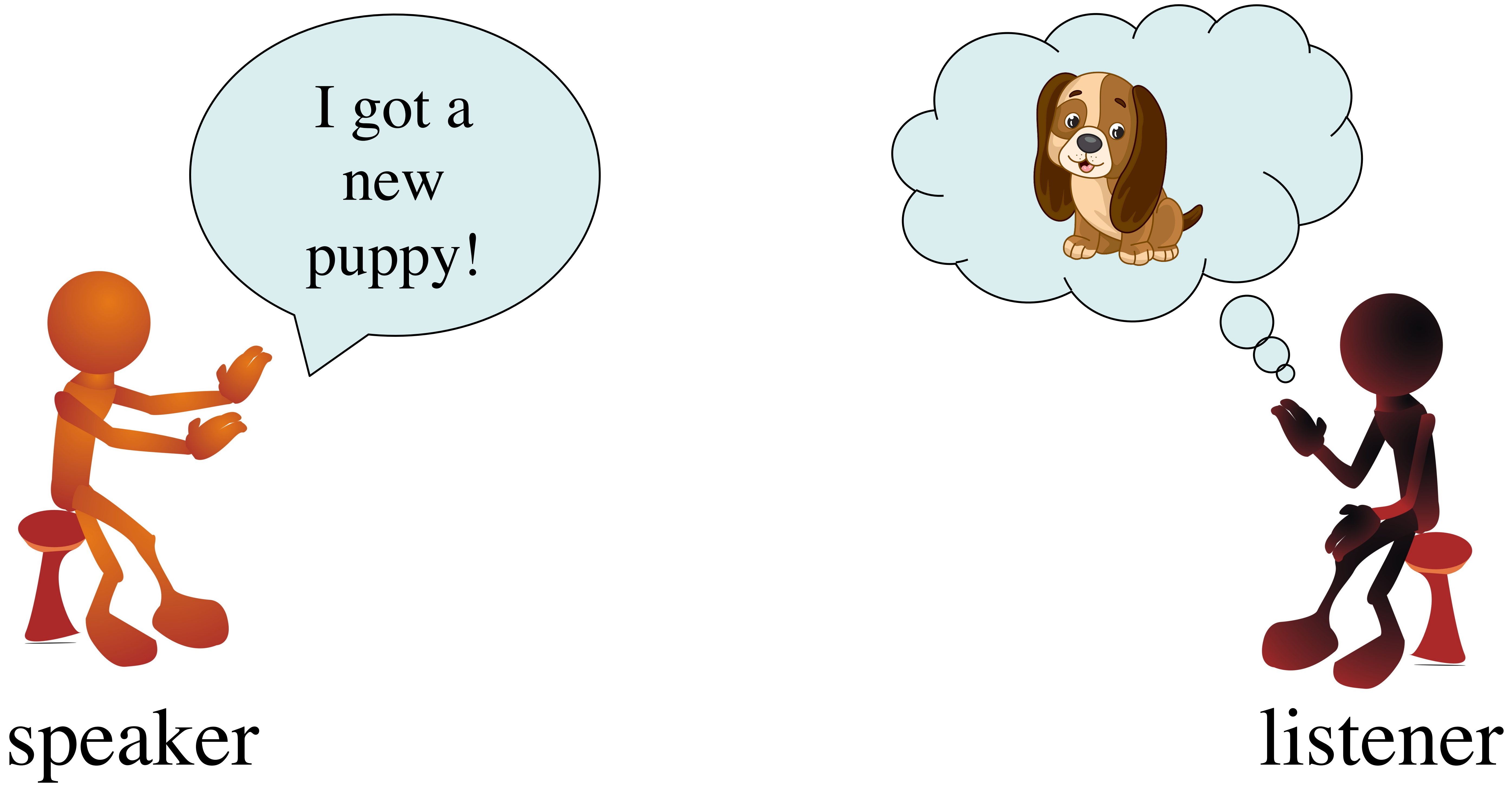


Preview for CMPT 983: Grounded natural language understanding

What is grounding?

Language is used to communicate about the world

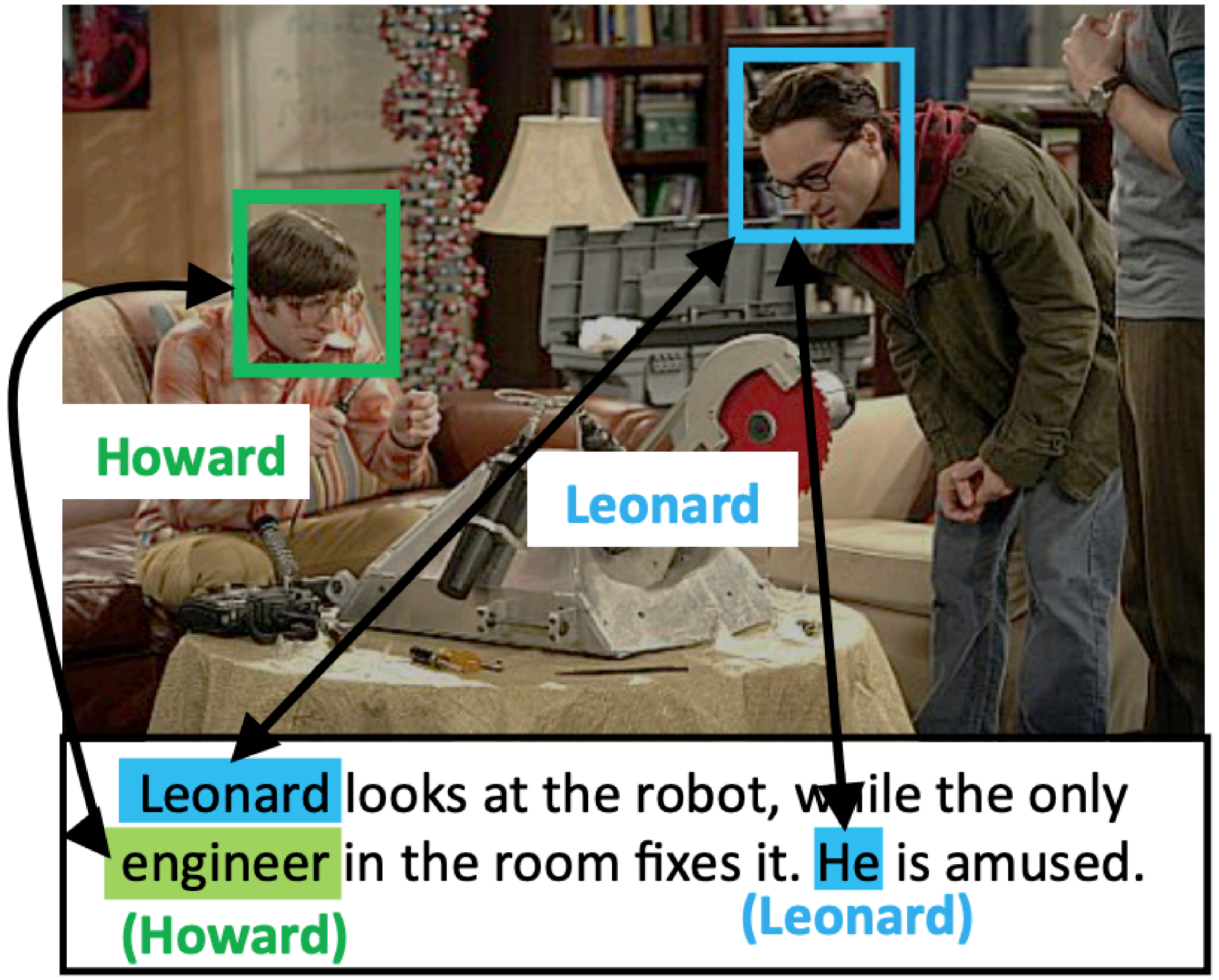
- Things, actions, abstract concepts



What is symbol grounding?

- Connecting **linguistic symbols** to their meaning

- Connecting words and sentences to what they represent

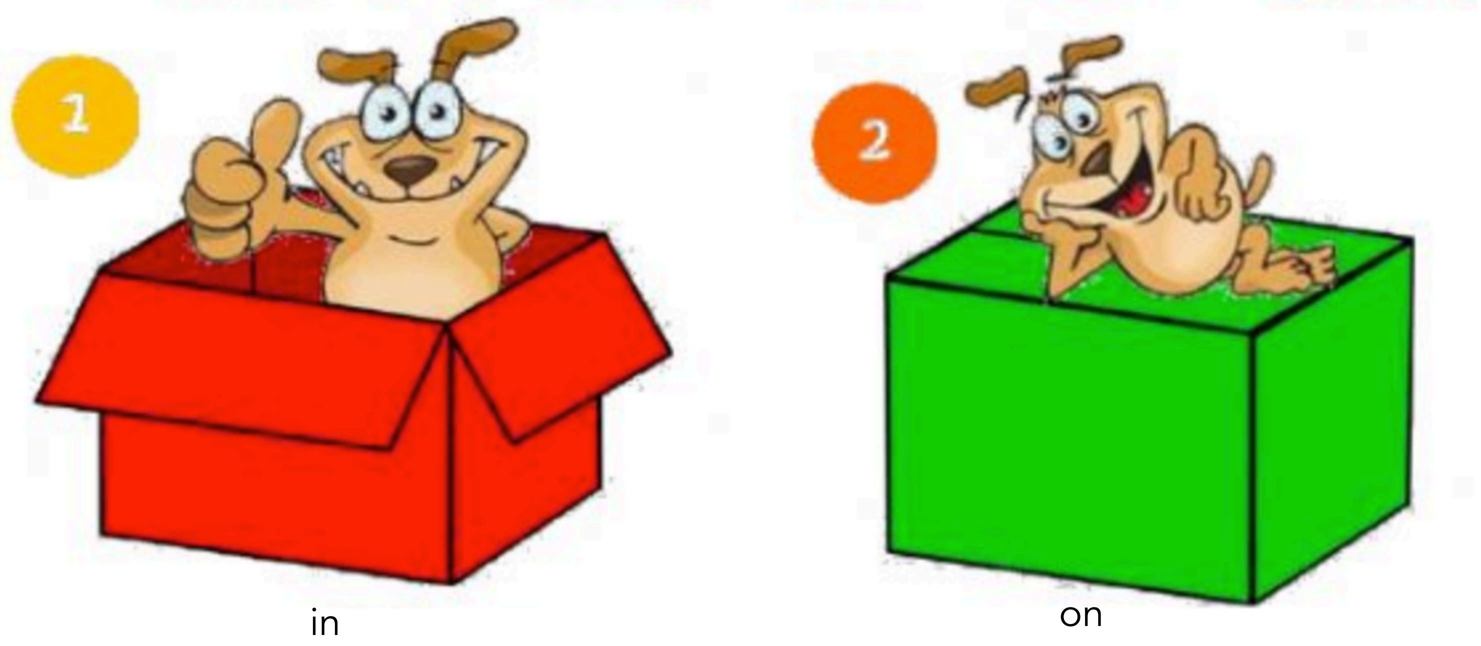


Linking people in videos with "their" names using coreference resolution [Ramanathan et al, 2014]

Actions

running

Spatial relations



Types of grounding

- **Perception**
  - Visual: *green* = [0,1,0] in RGB
  - Auditory: *loud* =   >120 dB
  - Taste: sweet = >some threshold level of sensation on taste buds
  - High-level concepts:

cat

dog

Slide adapted from Danqi Chen and Karthik Narasimhan

Types of grounding

- **Temporal concepts**
  - *late evening* = after 6pm
  - *fast, slow* = describing rates of change
- **Actions**

running

eating

Slide adapted from Danqi Chen and Karthik Narasimhan

Learning to connect linguistic symbols to the physical world

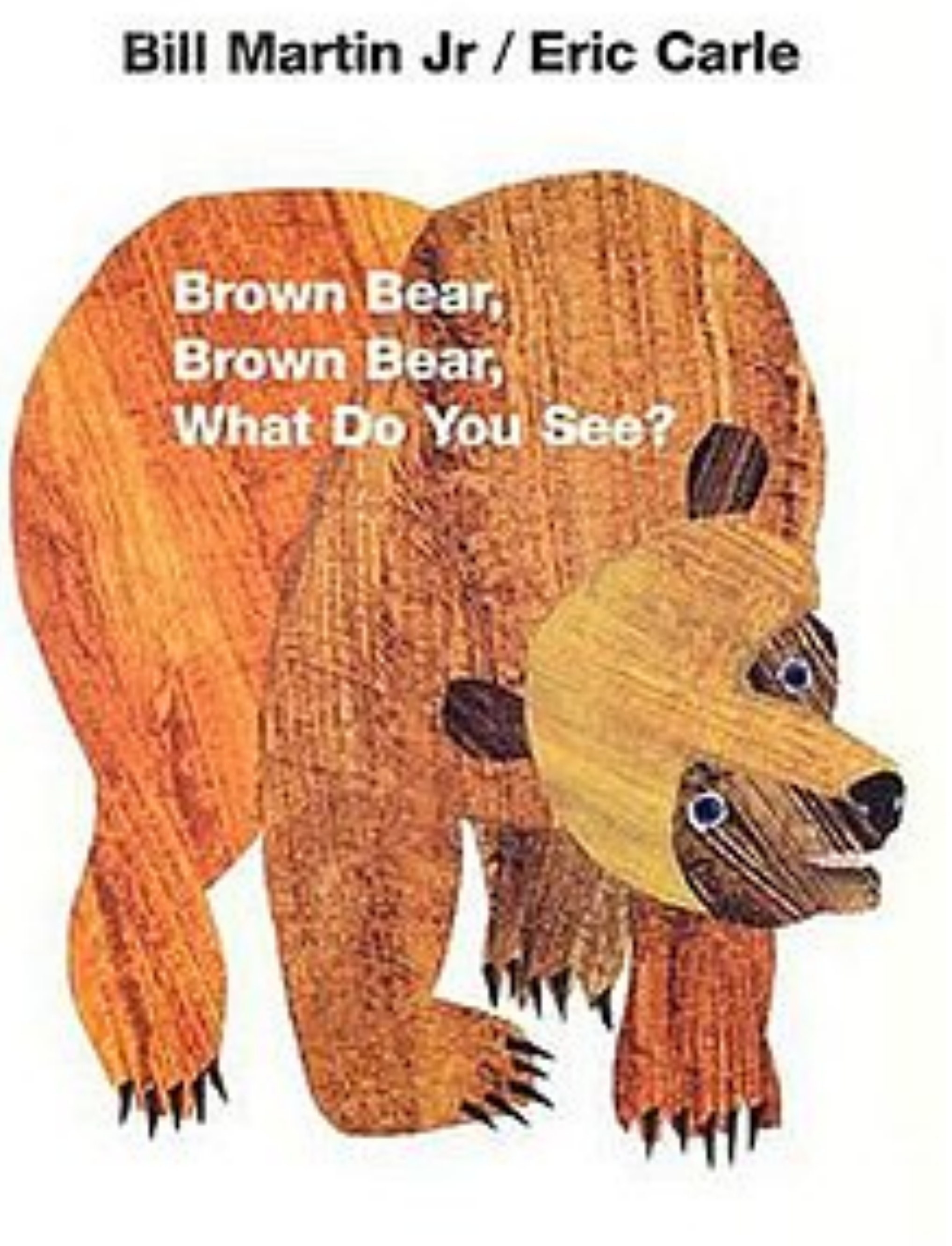
Children do not learn language from raw text
or passively watching TV

Natural way to learn language in the context of
its use in the physical and social world

This requires inferring the meaning of
utterances from their perceptual context

Children learn from multimodal sensory input and experience

Learning from multimodal information



Learn more about how children learn from Linda Smith: https://www.youtube.com/watch?v=dxli8qWJHLU

Choices in what to ground to

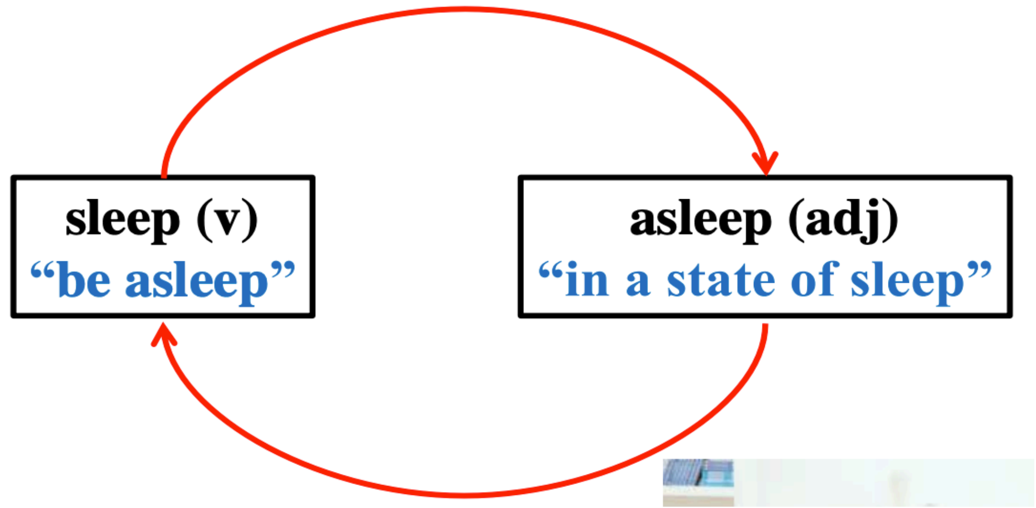
Connecting linguistic symbols to

- perceptual experiences and actions

Circular definitions

``Sleep" means ``be asleep"

sleep(n): ``a natural and periodic state of rest during which consciousness of the world is suspended"



One hundred → 100

- other symbols

The Big Bang Theory → https://en.wikipedia.org/wiki/The_Big_Bang_Theory

- to executable programs

Create a key `key` if it does not exist in dict `dic` and append element `value` to value

```
dic.setdefault(key, []).append(value)
```

Meaning representations

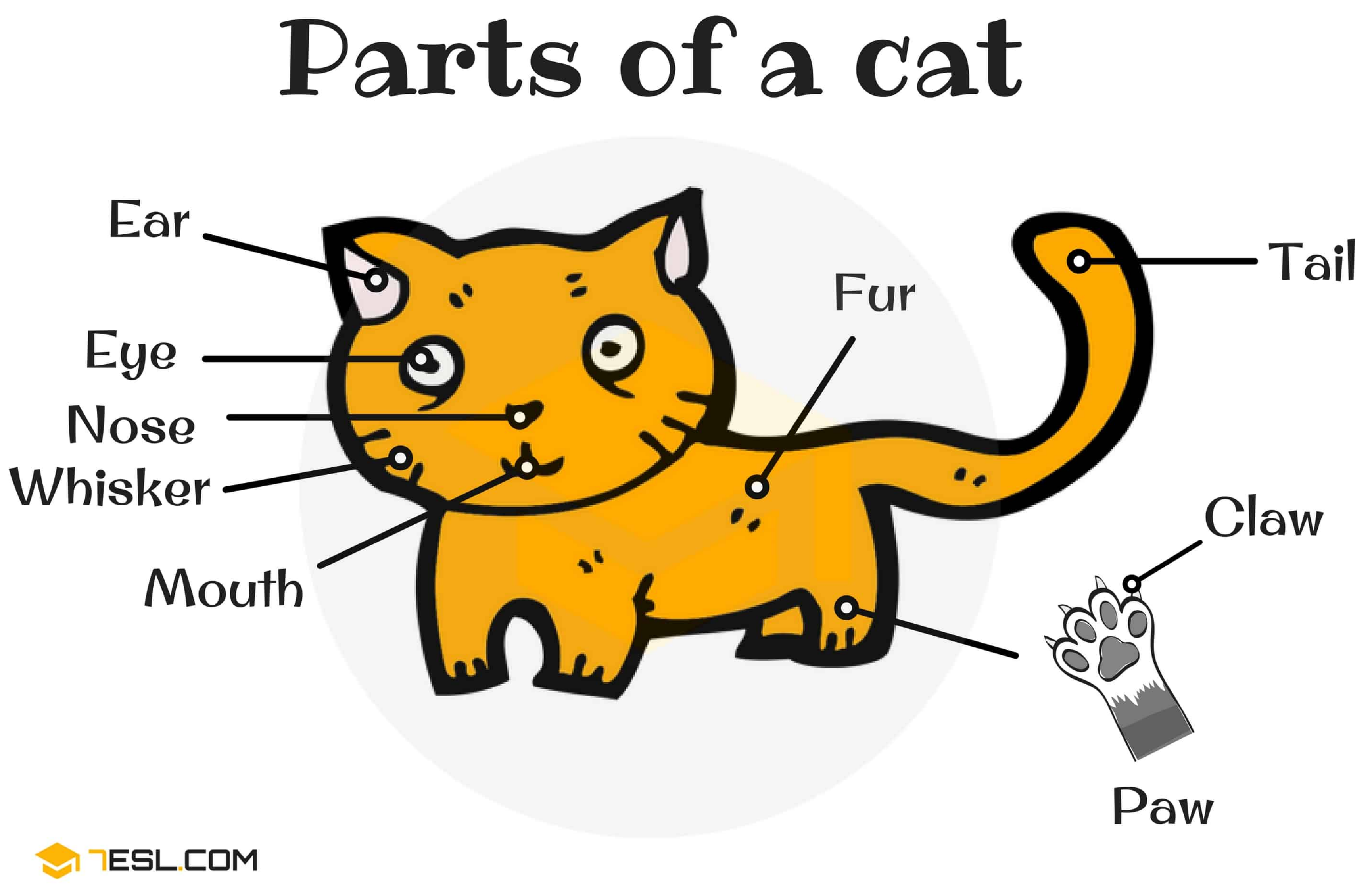
How do we represent the meaning of something?

"cat"

cat: a small domesticated carnivore, *Felis domestica* or *F. catus,* bred in a number of varieties.

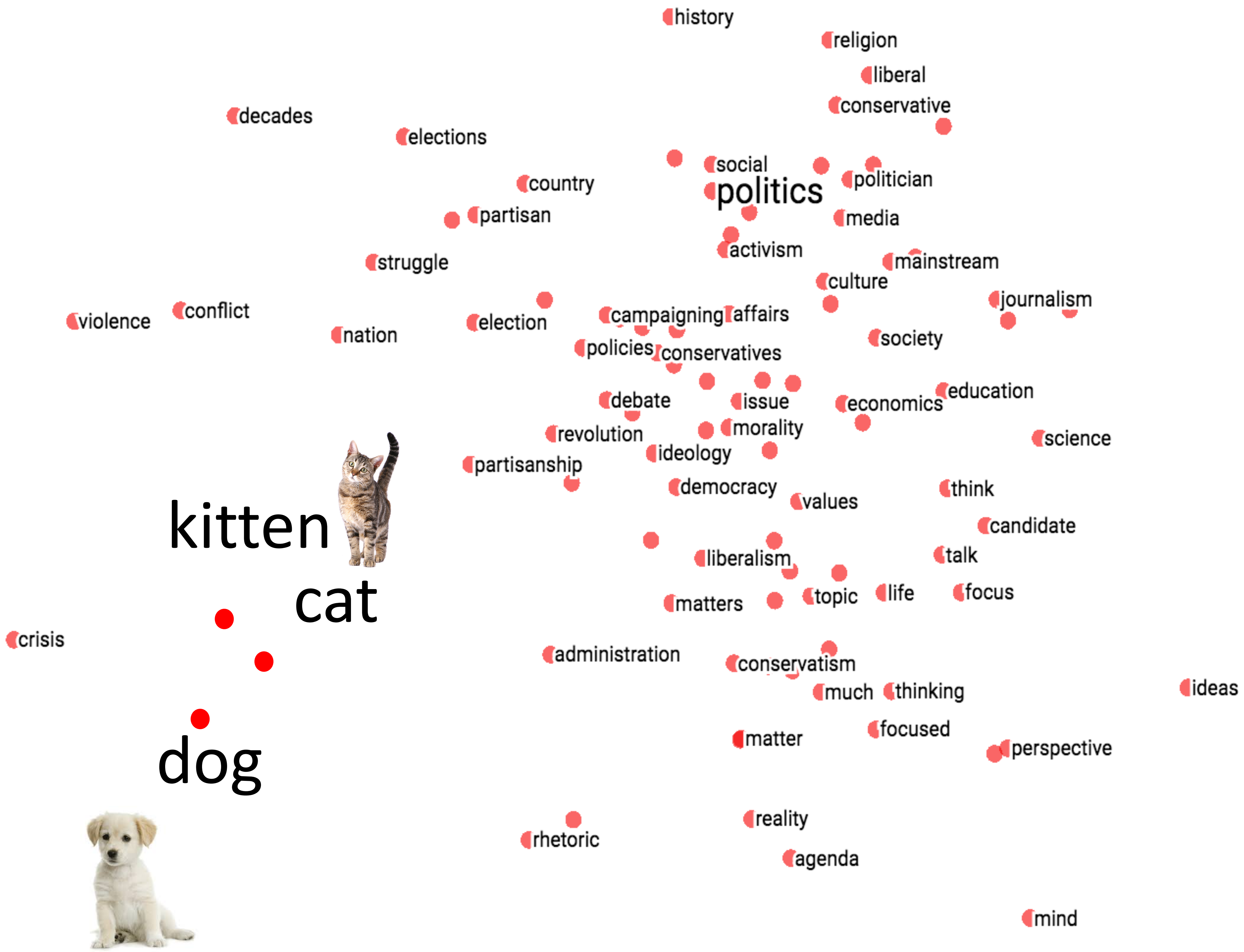
cat → {
  isMammal: true
  hasFur: true
  hasLegs: true
  meows: true
  barks: false
  height: 9.1 – 9.8 in
  weight: 7.9 – 9.9 lbs
  …
}

Attributed representation



Representations

Similar words closer to each other



Representing meaning as vectors

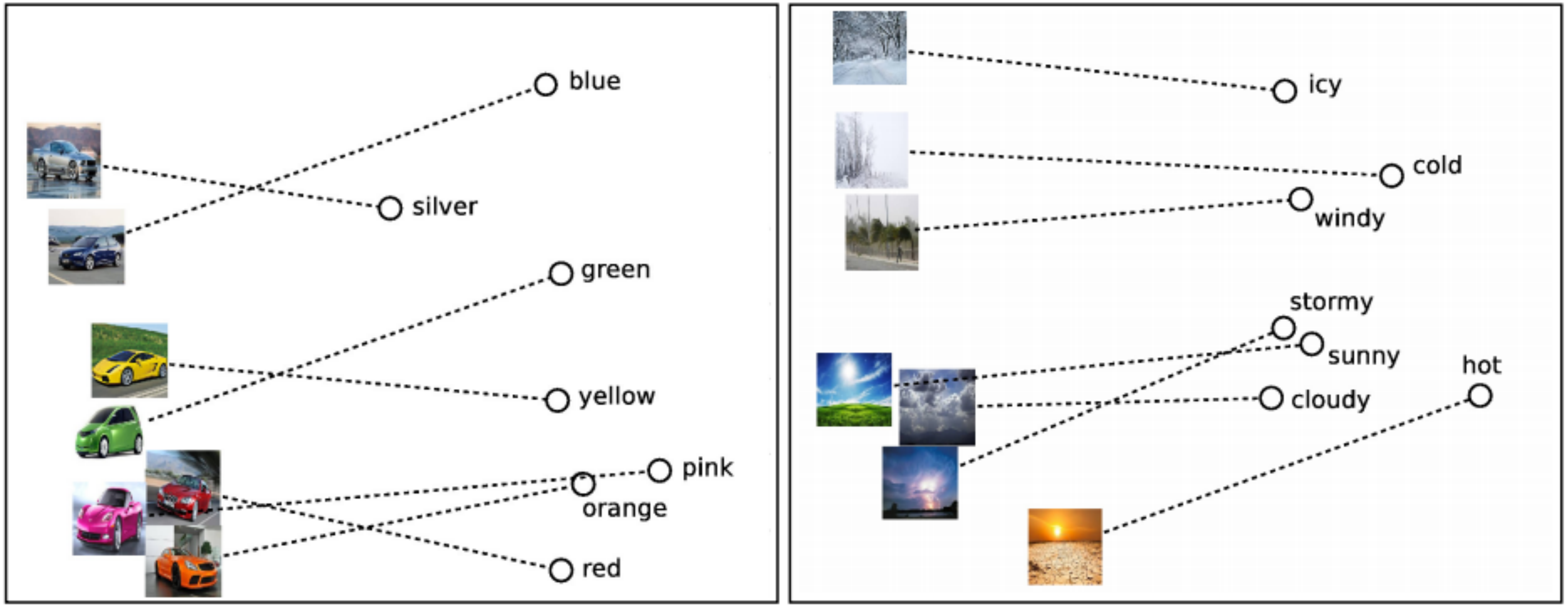
- common representation space
- enables information sharing
- can be learned from data

Embeddings in continuous vector space

cat = [0.04 1.79 -1.79 1.07 0.48]

dog = [0.61 1.84 -1.12 0.52 0.53]

Multimodal Embeddings



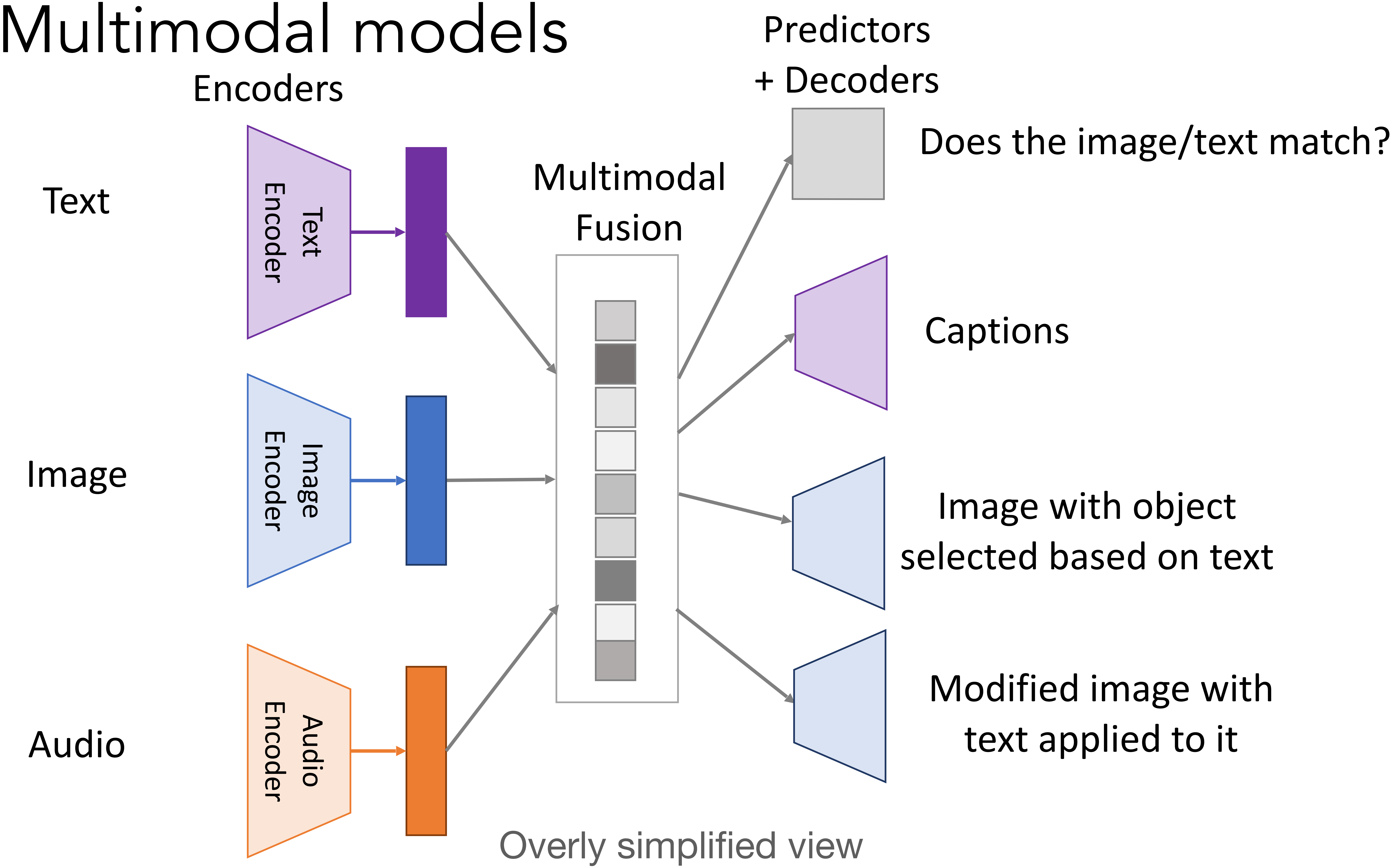
(a) Colors (b) Weather

Figure 5: PCA projection of the 300-dimensional word and image representations for (a) cars and colors and (b) weather and temperature.

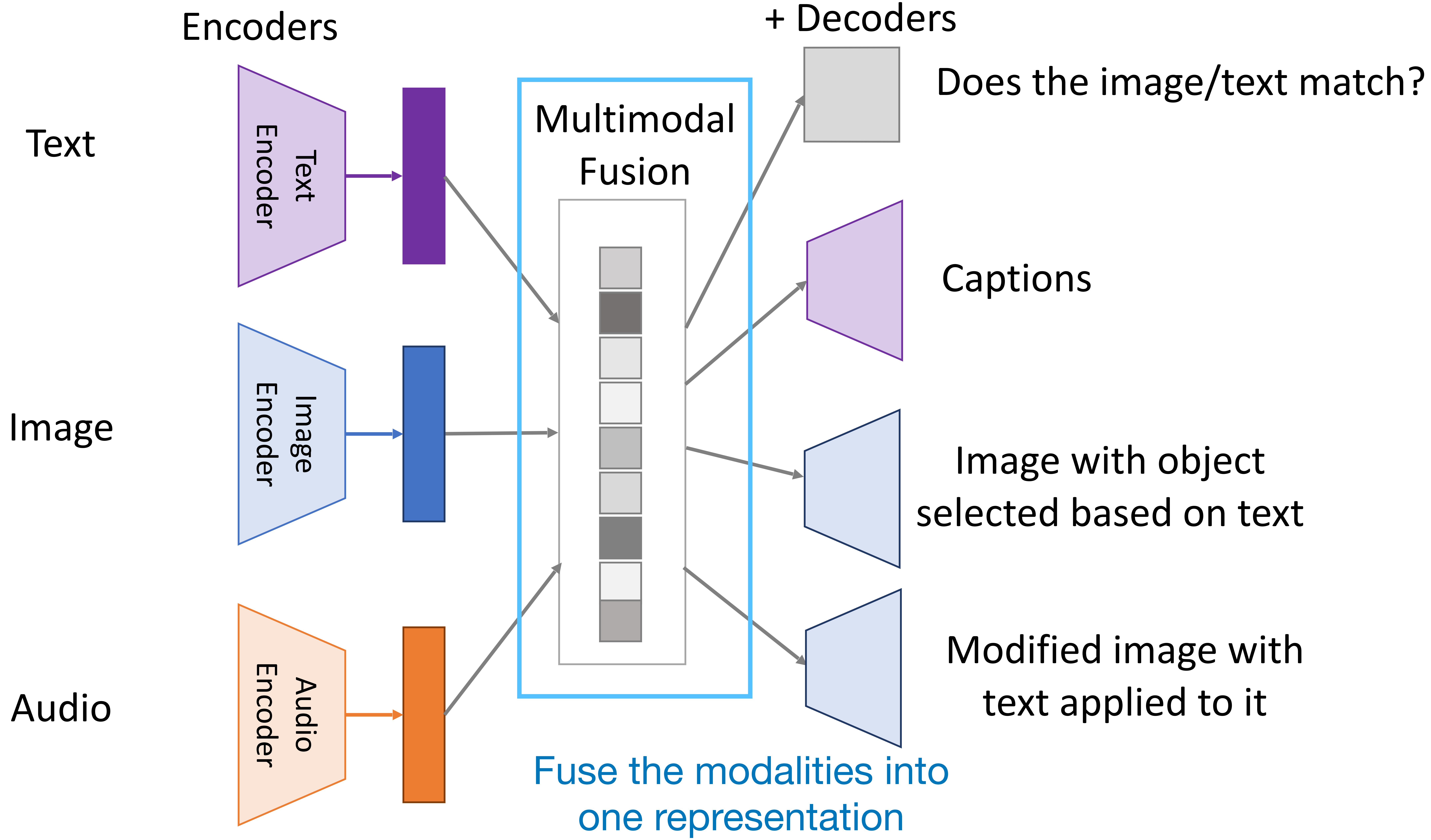
"Unifying Visual-Semantic Embeddings with Multimodal Neural Language Models"
[Kiros, Salakhutdinov, Zemel TACL 2015]

Cross-modal models

Multimodal models

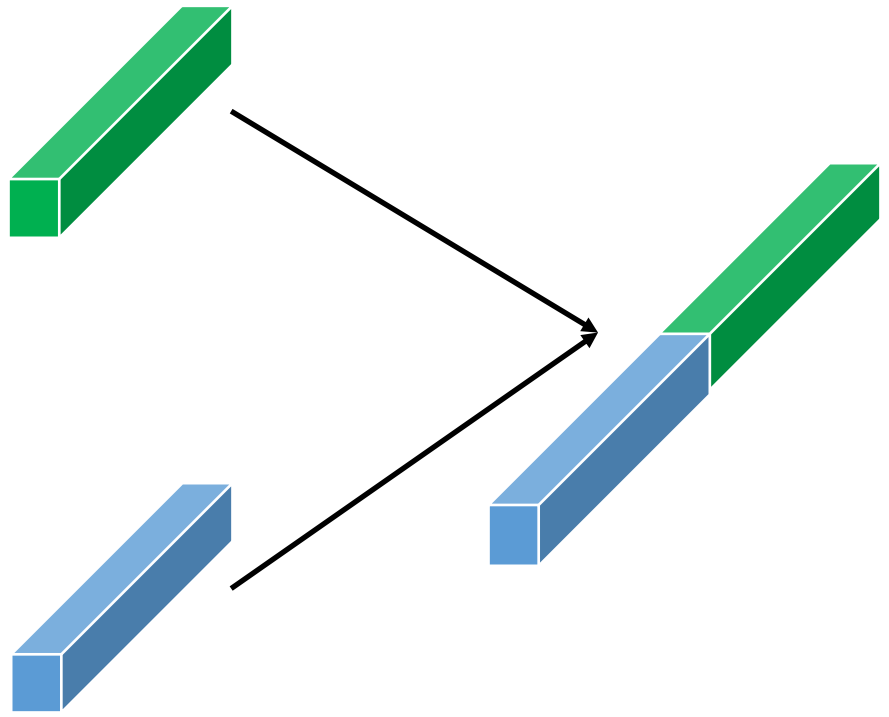


Multimodal models



Multimodal Fusion

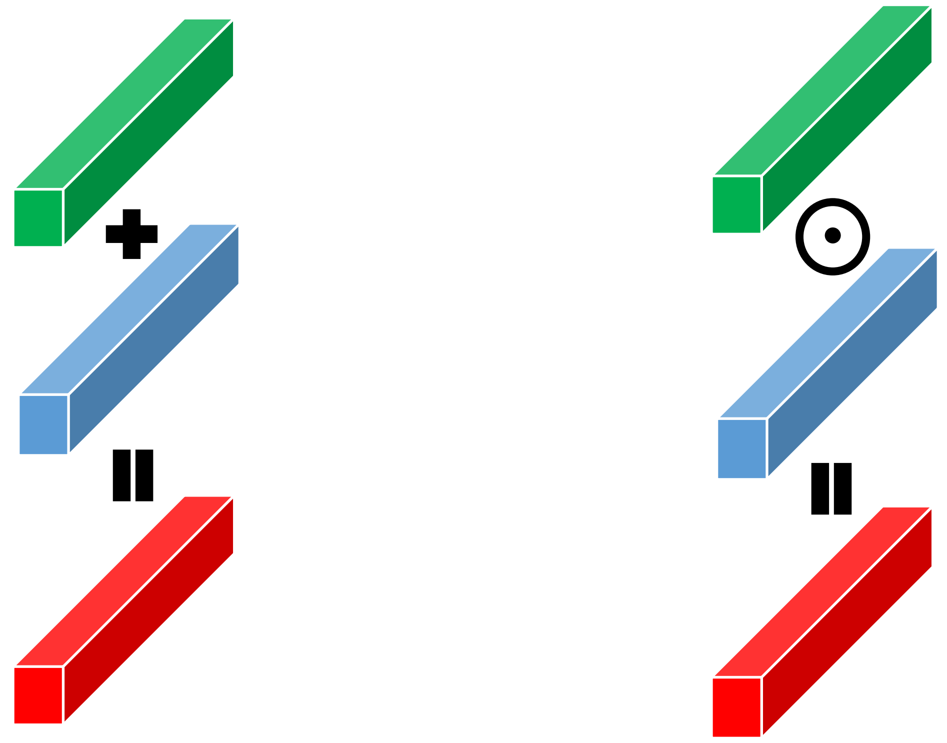
Concatenation



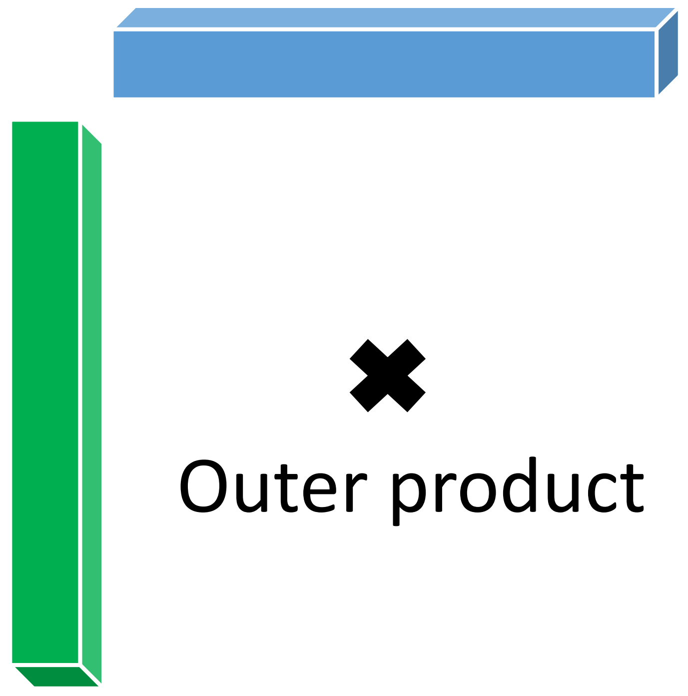
Element wise

Sum

Product



Bilinear Pooling



$$z = W\,[x \otimes q]$$

3000 2048 2048

12.5 billion !!!

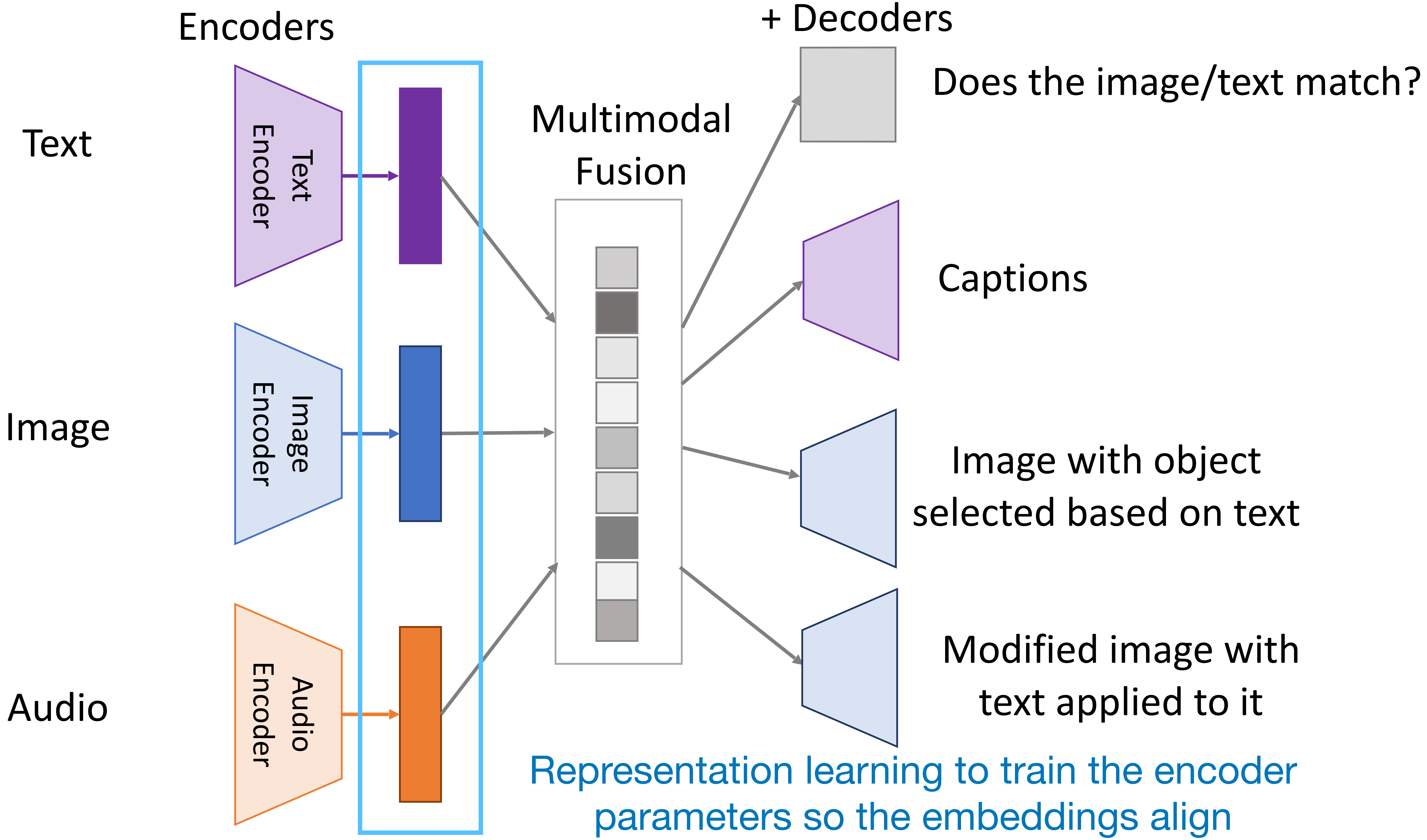
All elements can interact.
More flexible, but lots of weights!

Attention-based fusion

Use transformer to fuse modalities using attention

Image credits: Qi Wu

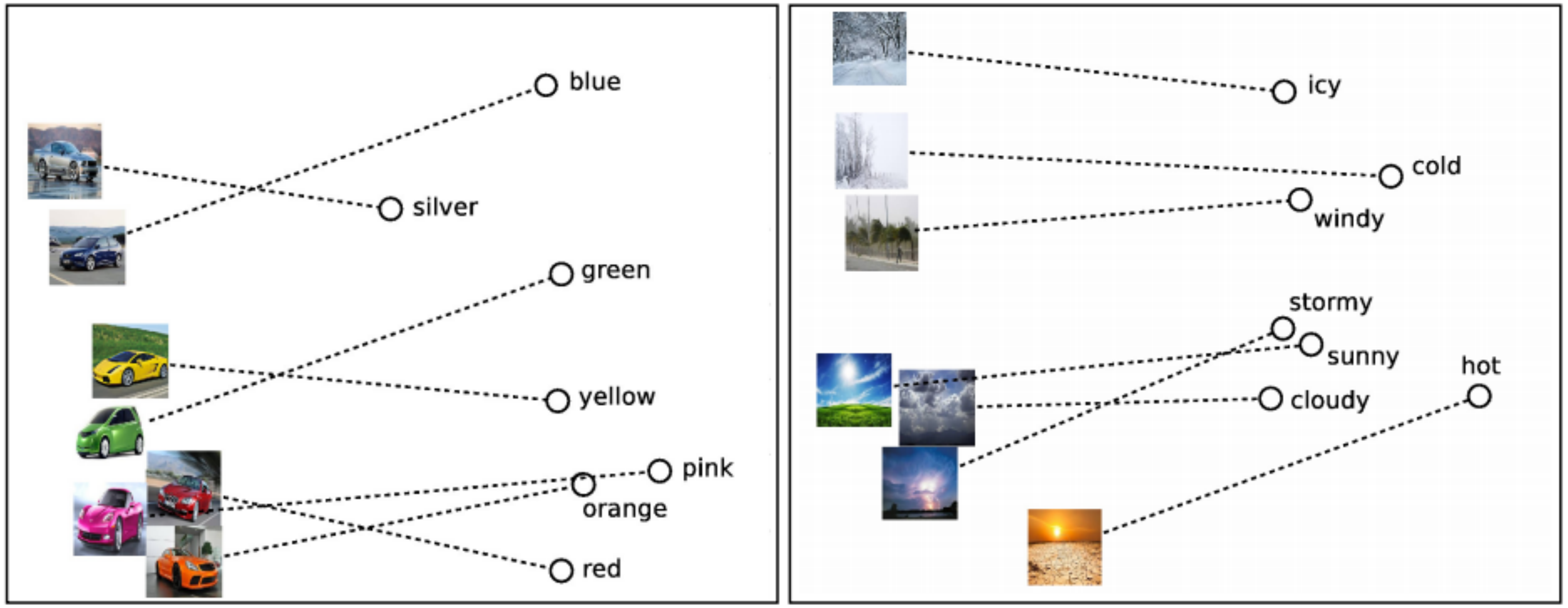
Multimodal models



Representation learning to train the encoder parameters so the embeddings align

Cross-modal Embeddings

Common representation for language and vision: vectors!



(a) Colors (b) Weather

Figure 5: PCA projection of the 300-dimensional word and image representations for (a) cars and colors and (b) weather and temperature.

Unifying Visual-Semantic Embeddings with Multimodal Neural Language Models
[Kiros, Salakhutdinov, Zemel TACL 2015, https://arxiv.org/pdf/1411.2539.pdf]

Cross-modal Embeddings

Images and **class labels** are embedded into the same space

Image Embedding

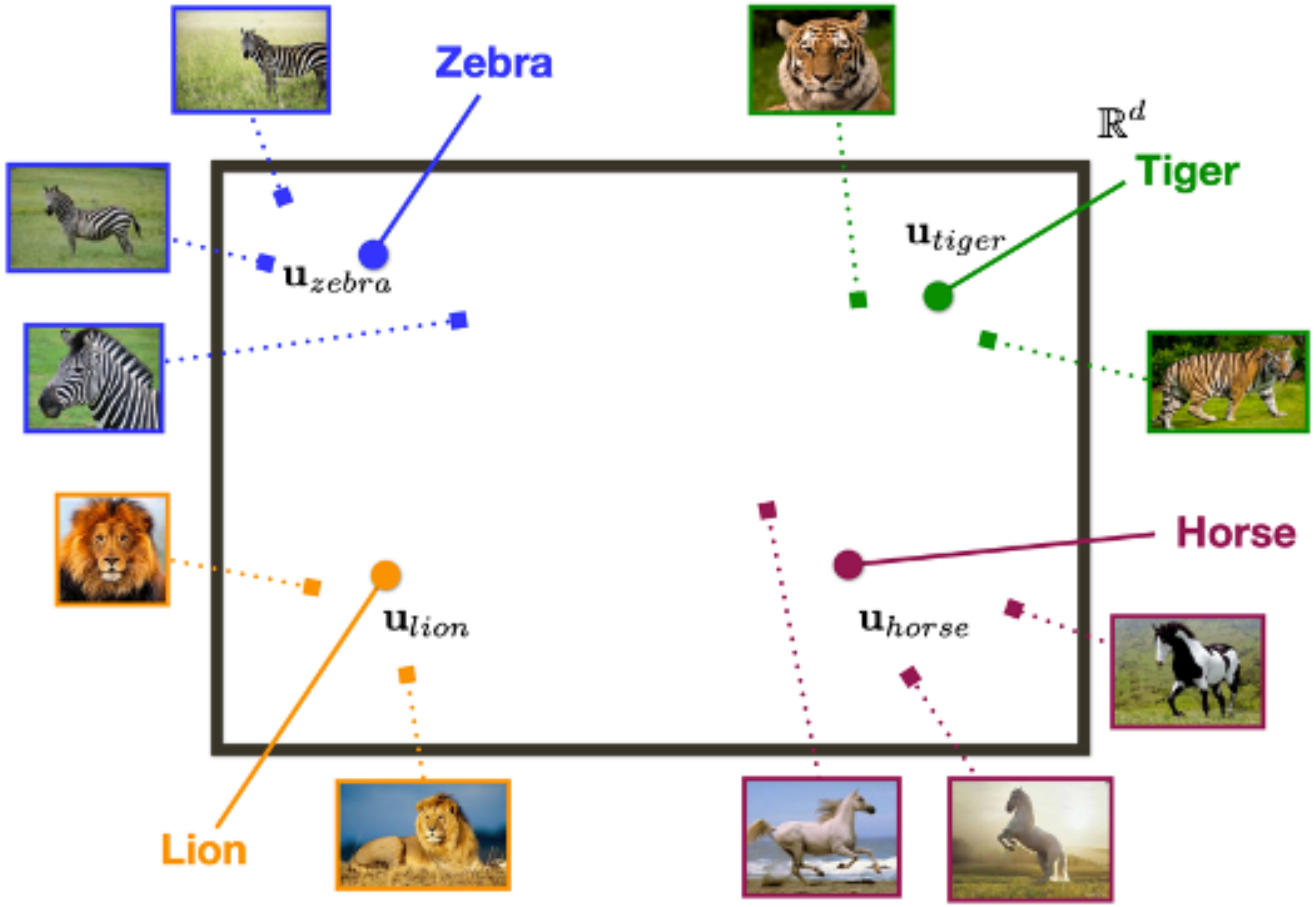
$$\Psi(I_i) = \mathbf{W} \cdot CNN(I_i; \mathbf{\Theta}) : \mathbb{R}^D \rightarrow \mathbb{R}^d$$

Label Embedding

$$\Psi_L(word_i) = \mathbf{u}_i : \{1, ..., L\} \rightarrow \mathbb{R}^d$$

Similarity in Embedding Space

$$S(\mathbf{u}, \mathbf{u}') = \frac{\mathbf{u}}{\|\mathbf{u}\|} \cdot \frac{\mathbf{u}'}{\|\mathbf{u}'\|}$$



Adapted from slide by Leonid Sigal

Cross-modal Embeddings

Correct label (more similar)

Other labels (less similar)

$$\mathcal{L}_C = \sum \max(0, \alpha - \boxed{S(\Psi(I_i), \mathbf{u}_{y_i})} + \boxed{S(\Psi(I_i), \mathbf{u}_{y_c})})$$

+ pair

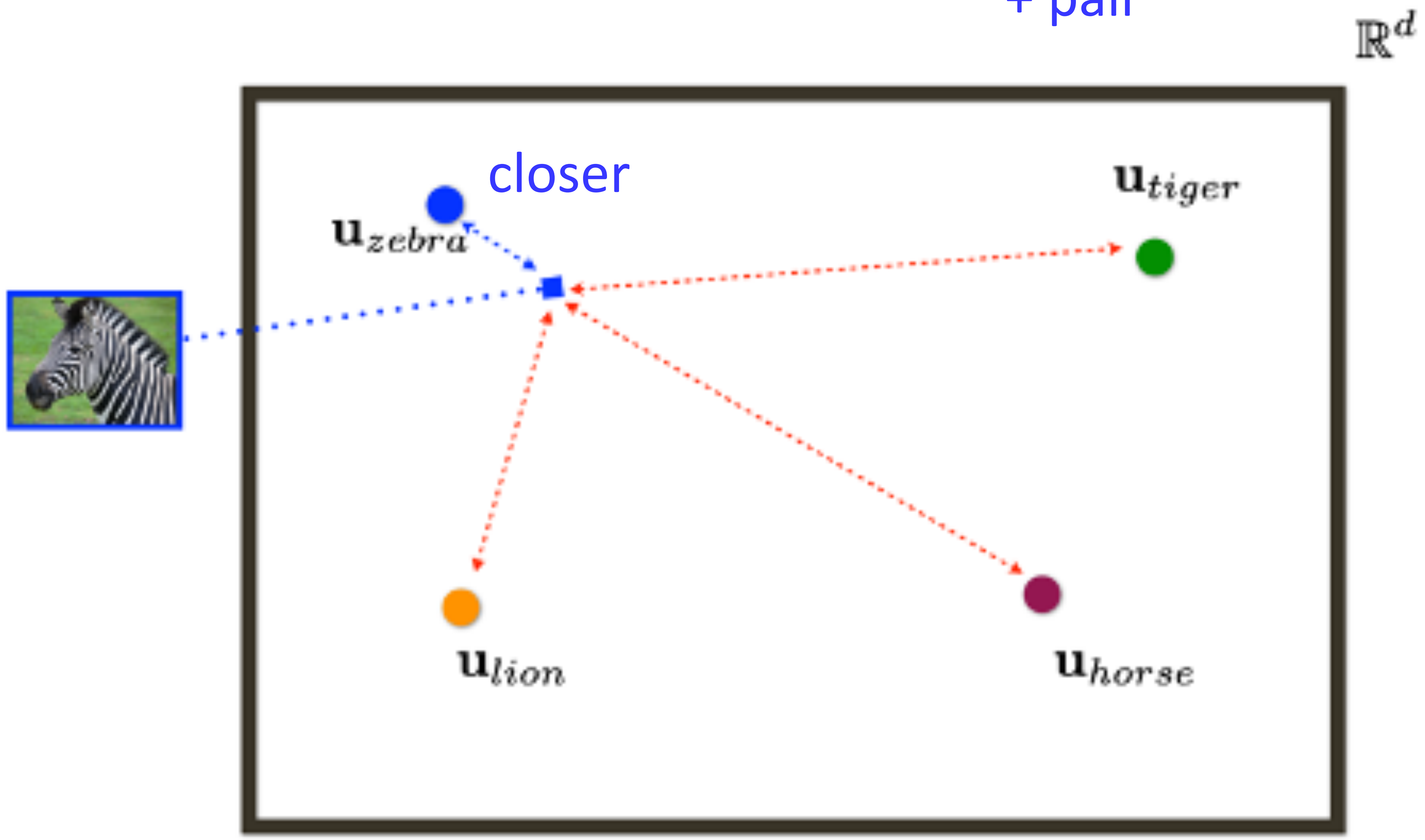
- pair

Image Embedding

$\Psi(I_i) = \mathbf{W} \cdot CNN(I_i; \mathbf{\Theta}) \colon \mathbb{R}^D \rightarrow \mathbb{R}^d$

Label Embedding

$\Psi_L(word_i) = \mathbf{u}_i : \{1, ..., L\} \rightarrow \mathbb{R}^d$



Similarity in Embedding Space

$$S(\mathbf{u}, \mathbf{u}') = \frac{\mathbf{u}}{\|\mathbf{u}\|} \cdot \frac{\mathbf{u}'}{\|\mathbf{u}'\|}$$

Objective Function:

$$\min_{\mathbf{W},\mathbf{U}} \sum_i^N \mathcal{L}_C(\mathbf{W}, \mathbf{U}, I_i, y_i) + \lambda_1 ||\mathbf{W}||_F^2 + \lambda_2 ||\mathbf{U}||_F^2$$

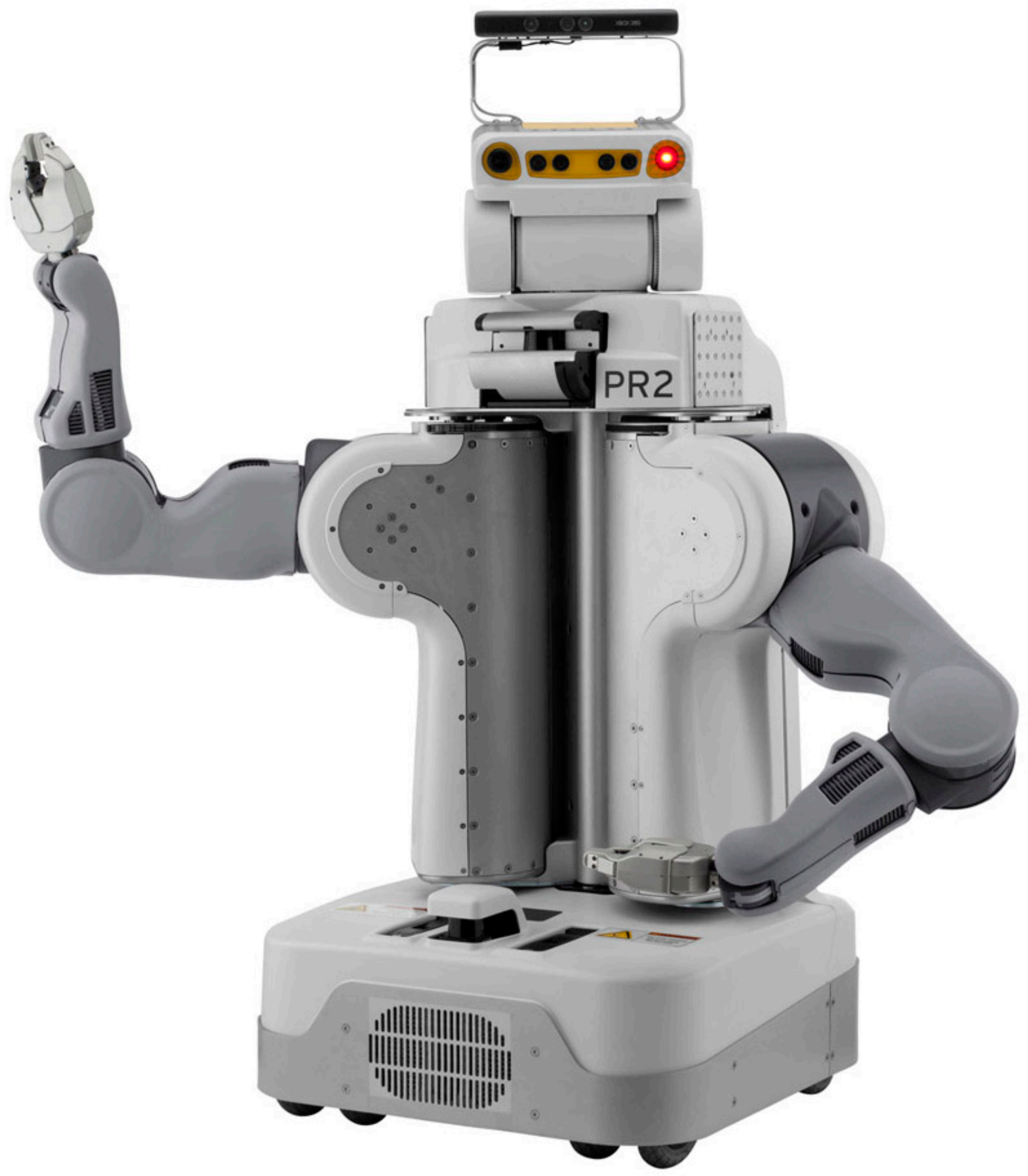
[Bengio *et al.*,, NIPS'10]

[Weinberger, Chapelle, NIPS'09]

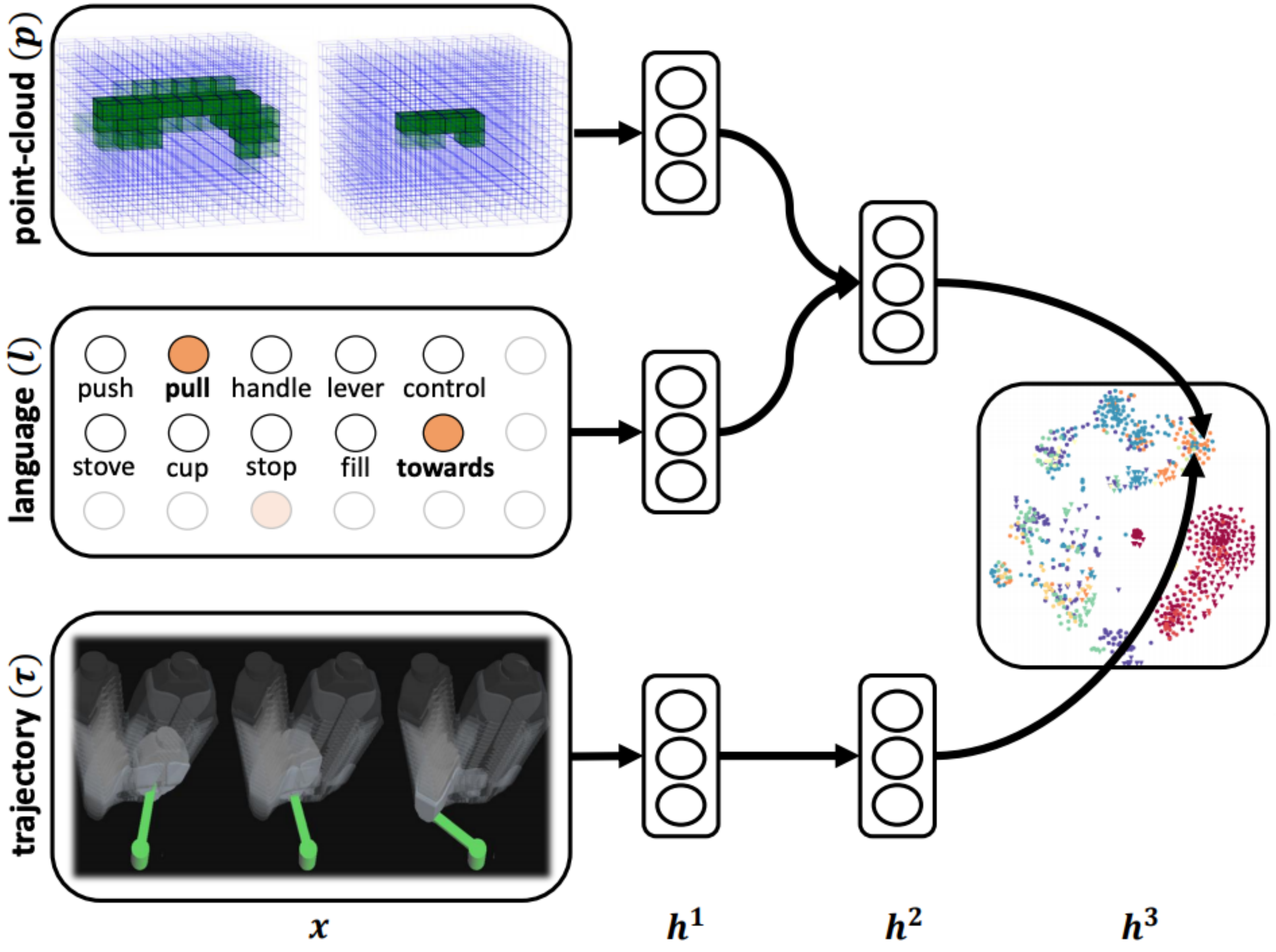
Adapted from slide by Leonid Sigal

Can embed anything!

PR2 robot



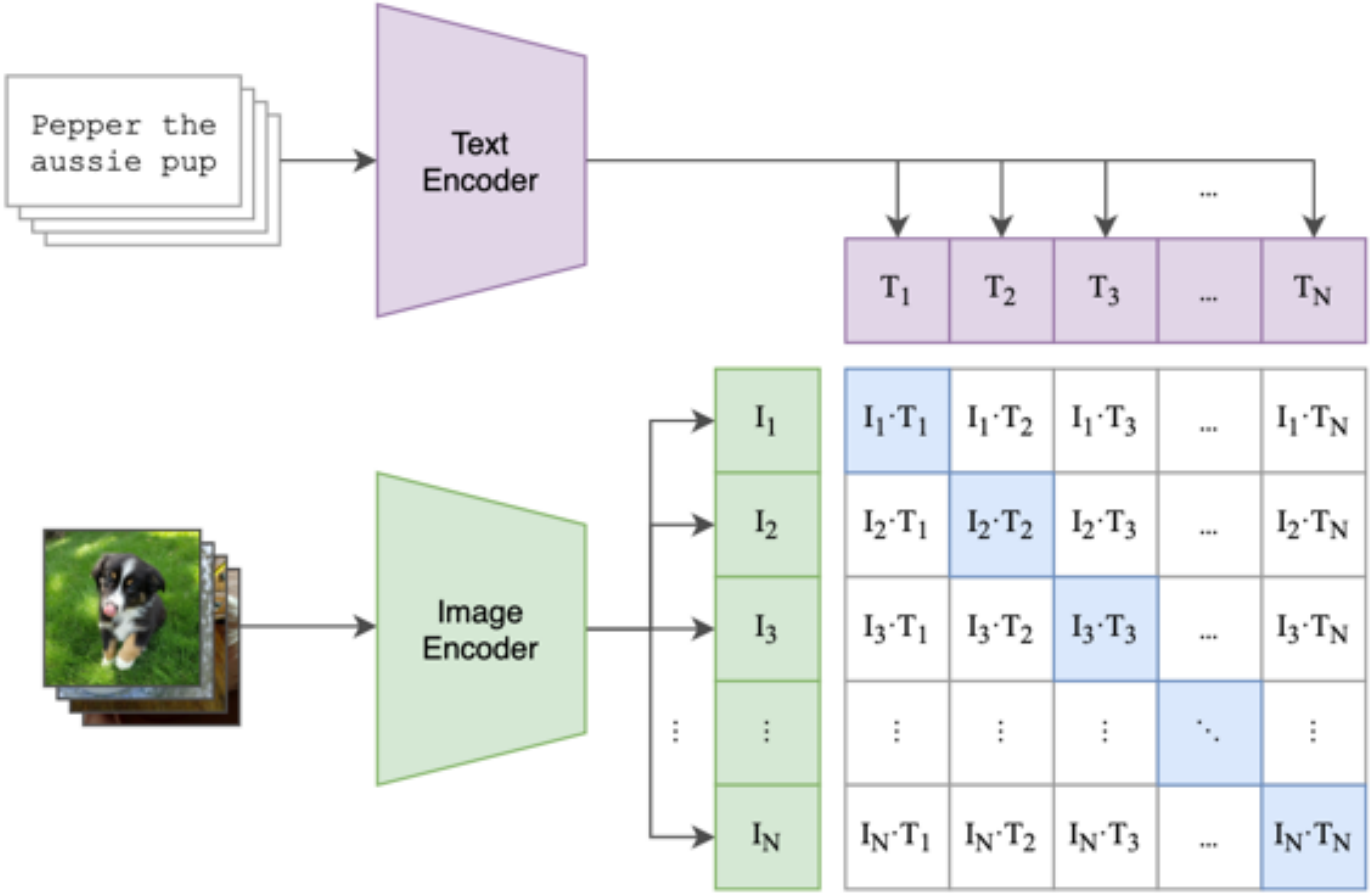
https://robots.ieee.org/robots/pr2



Deep Multimodal Embedding: Manipulating Novel Objects with Point-clouds, Language and Trajectories, Sung et al, arxiv 2015

Contrastive pretraining

OpenAI CLIP

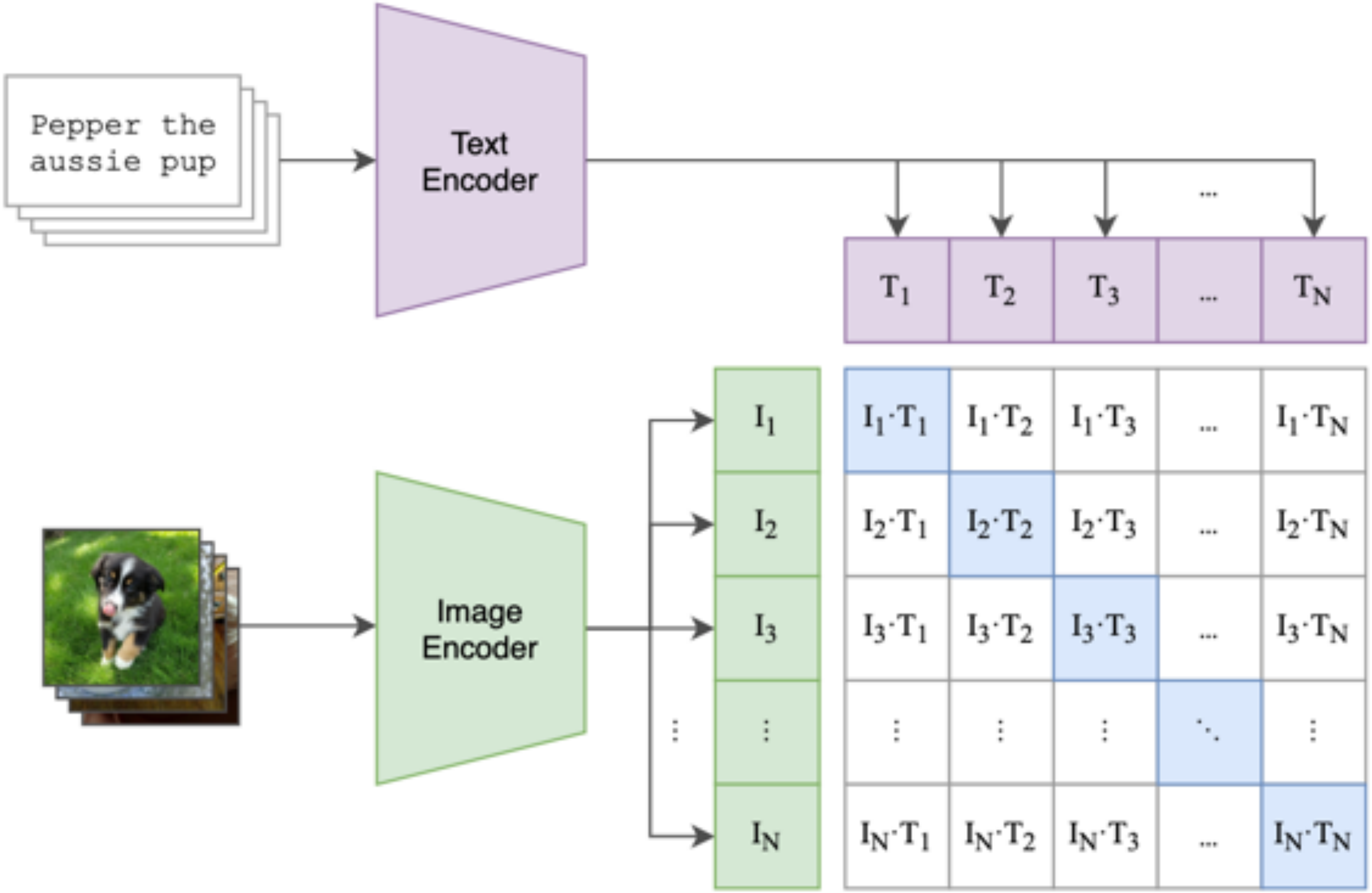


- Train on large amount of data
  - WebImageText: 400M text-image pairs
- Contrastive pretraining: does the text-image pair match?
  - Batch size N=32K
  - N positive pairs
  - $N^2 - N$ negative pairs
- Transformer based model for both vision and language

Learning transferable visual models from natural from natural language supervision, Radford et al, 2020

Contrastive pretraining

OpenAI CLIP




```
# image_encoder - ResNet or Vision Transformer
# text_encoder  - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l]       - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t             - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T)  #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_i + loss_t)/2
```

Learning transferable visual models from natural from natural language supervision, Radford et al, 2020

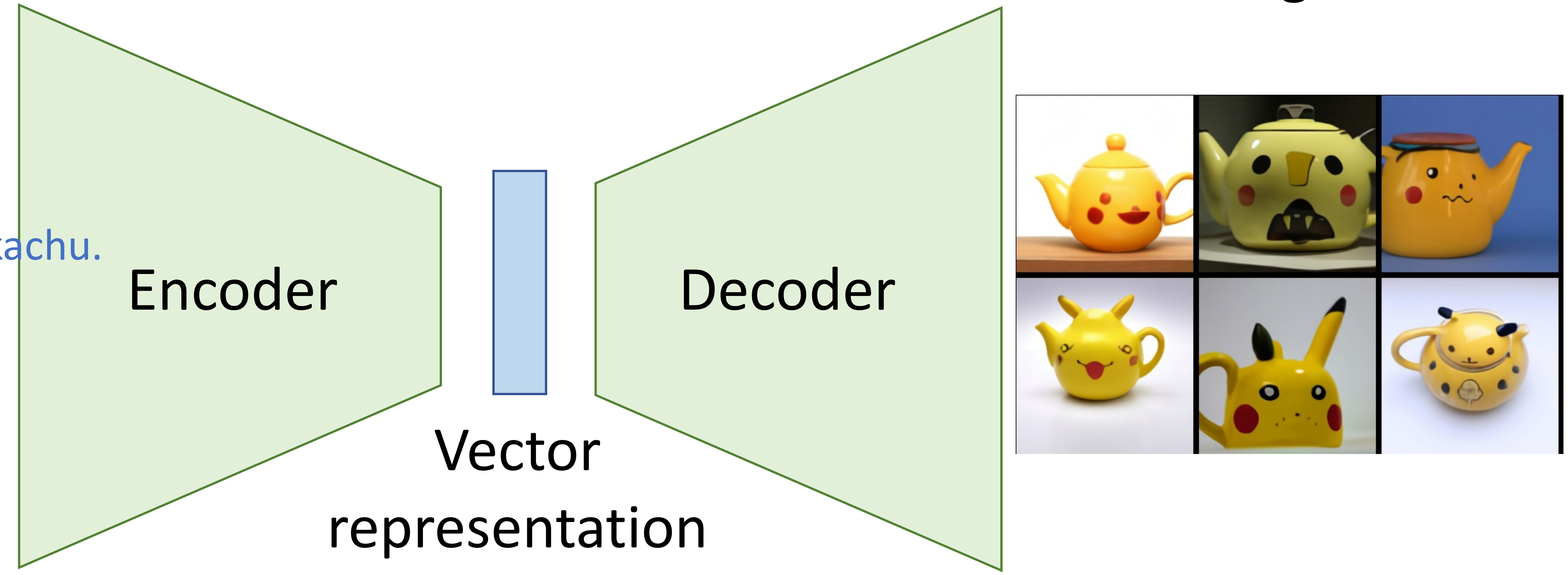
Used in DALL-E

To rerank generated captions: select 32 out of 512

Text

Image

a teapot in the shape of a pikachu.
a teapot imitating a pikachu

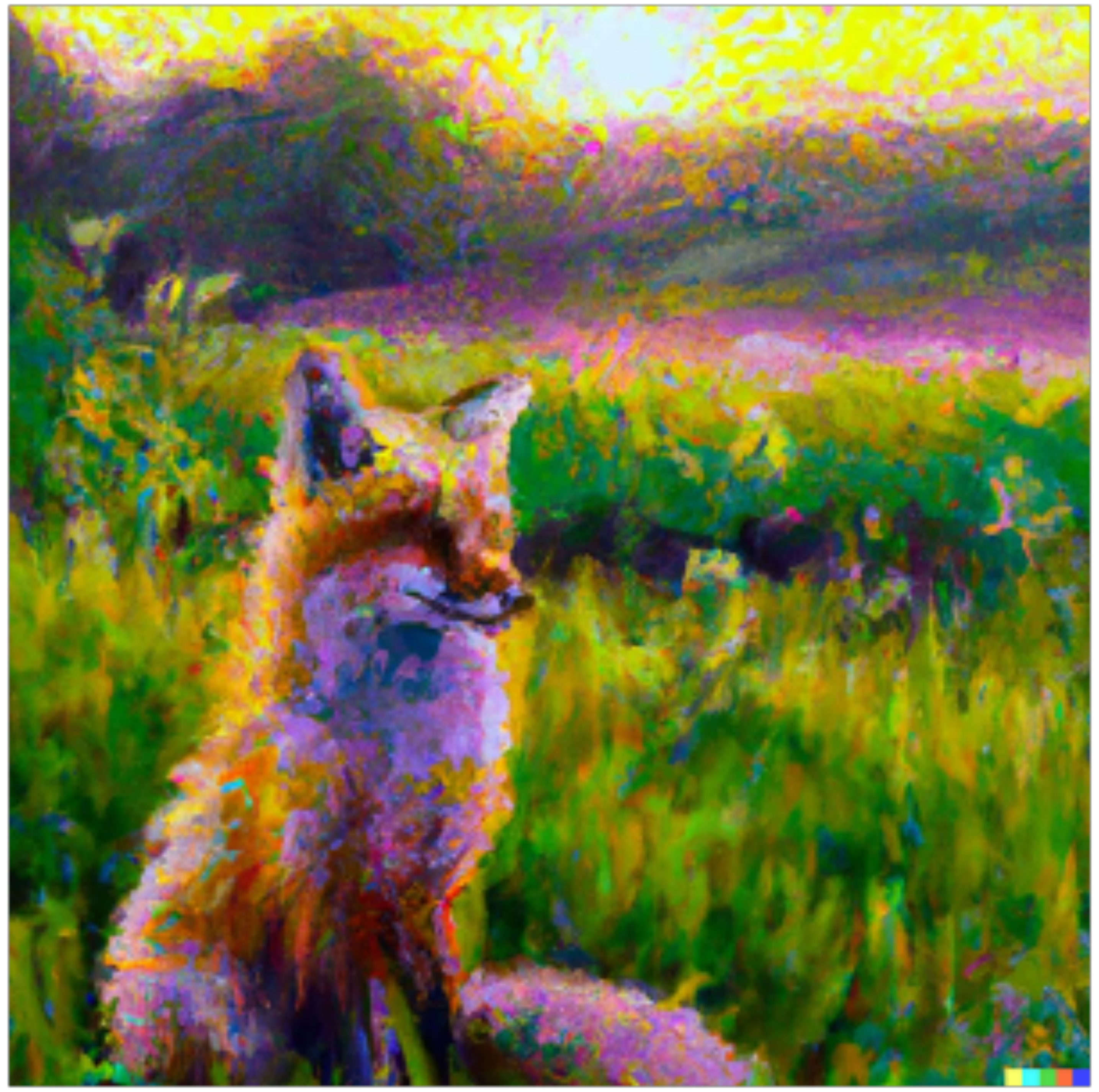


"Dall-e"
[Ramesh et al, https://openai.com/blog/dall-e/]

Text-to-Image Generation with Diffusion Models

OpenAI DALL-E 2

a painting of a fox sitting in a field at sunrise in the style of Claude Monet



https://openai.com/dall-e-2/

Google Imagen

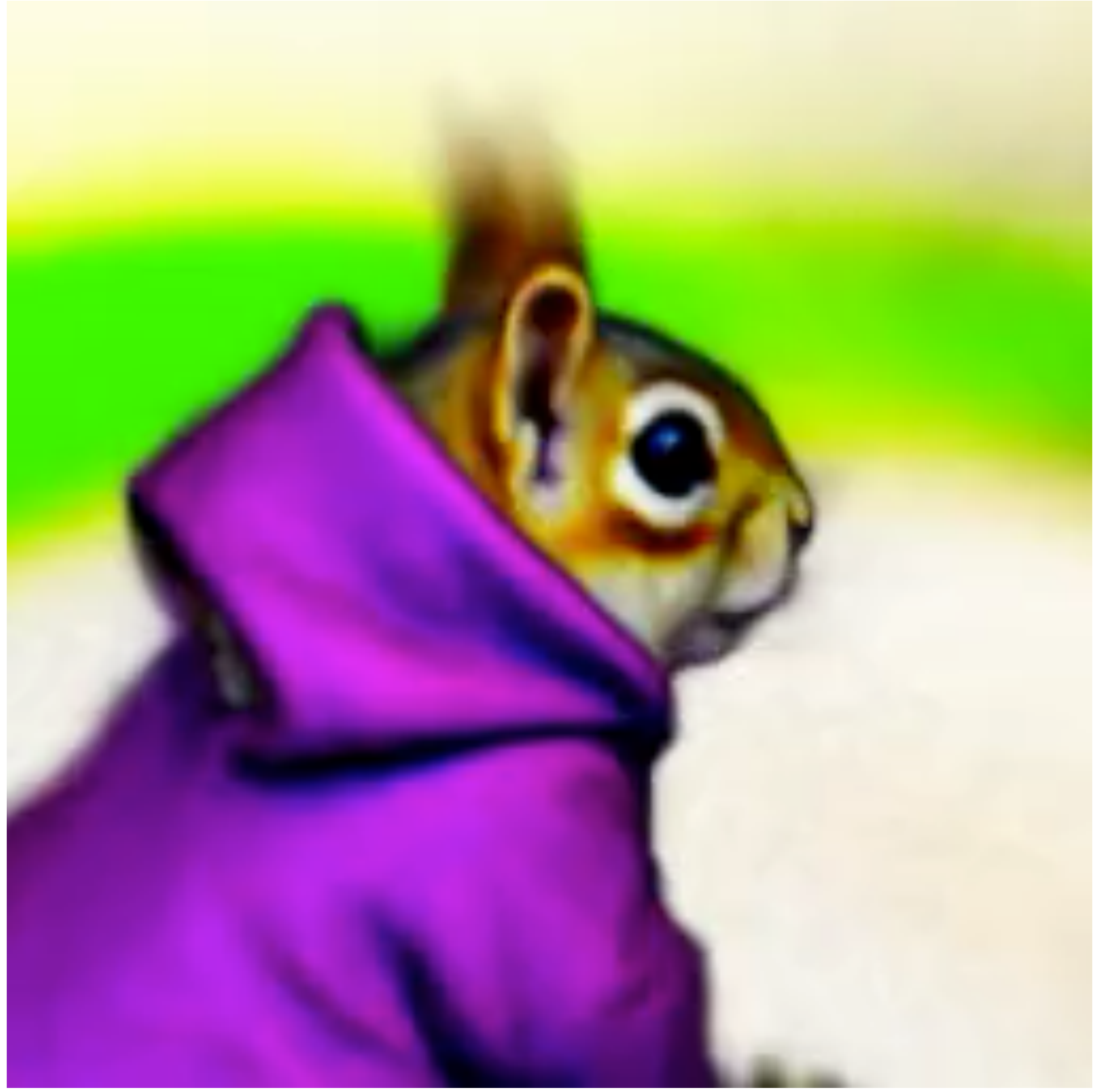
A cute sloth holding a small treasure chest. A bright golden glow is coming from the chest.

https://imagen.research.google/

Try it out yourself: https://huggingface.co/spaces/stabilityai/stable-diffusion, https://www.craiyon.com/, https://www.midjourney.com/home/

Text-to-3D with diffusion models

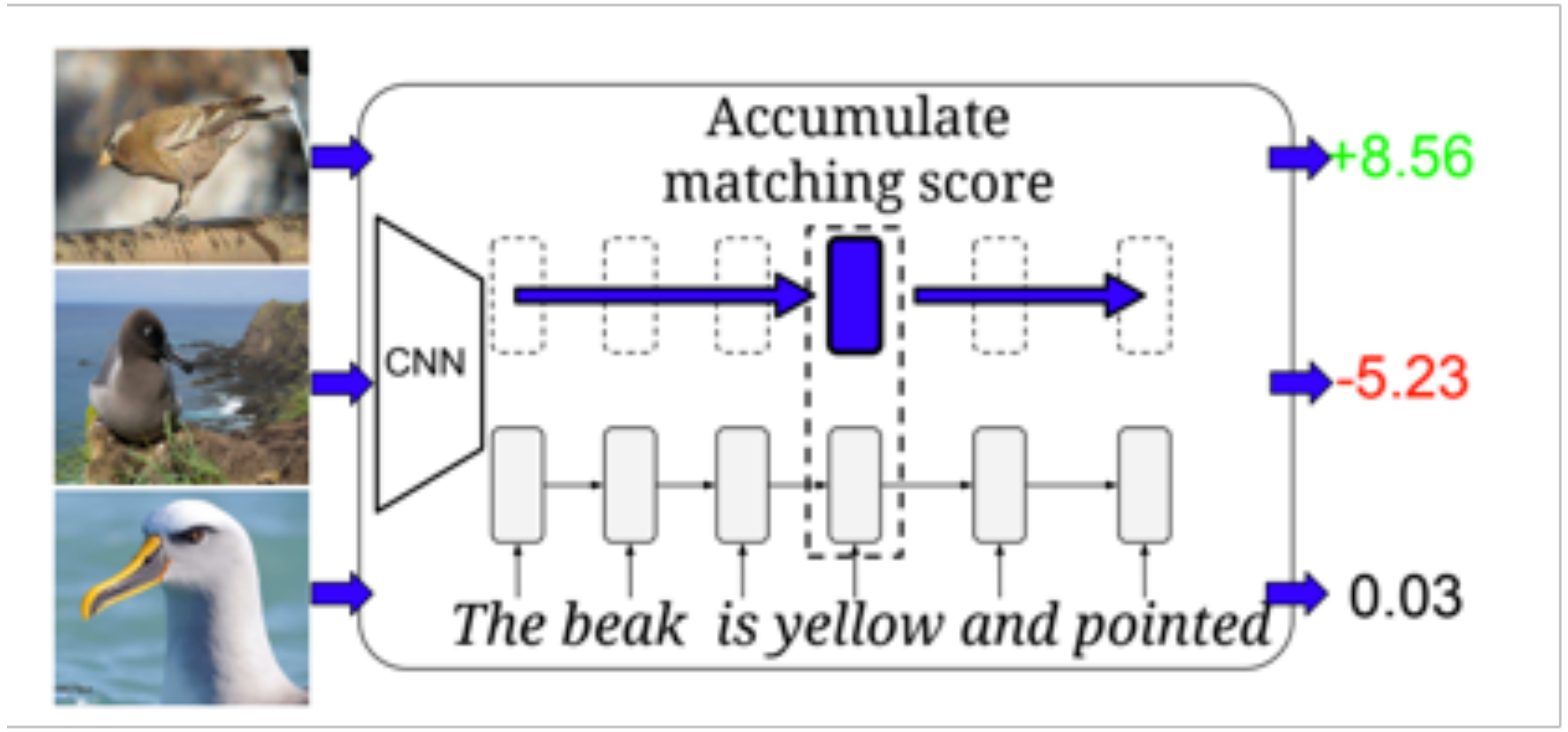
a DSLR photo of a squirrel wearing a purple hoodie



https://dreamfusionpaper.github.io/

Retrieval

- Text to image/video retrieval
- Image/video to text retrieval



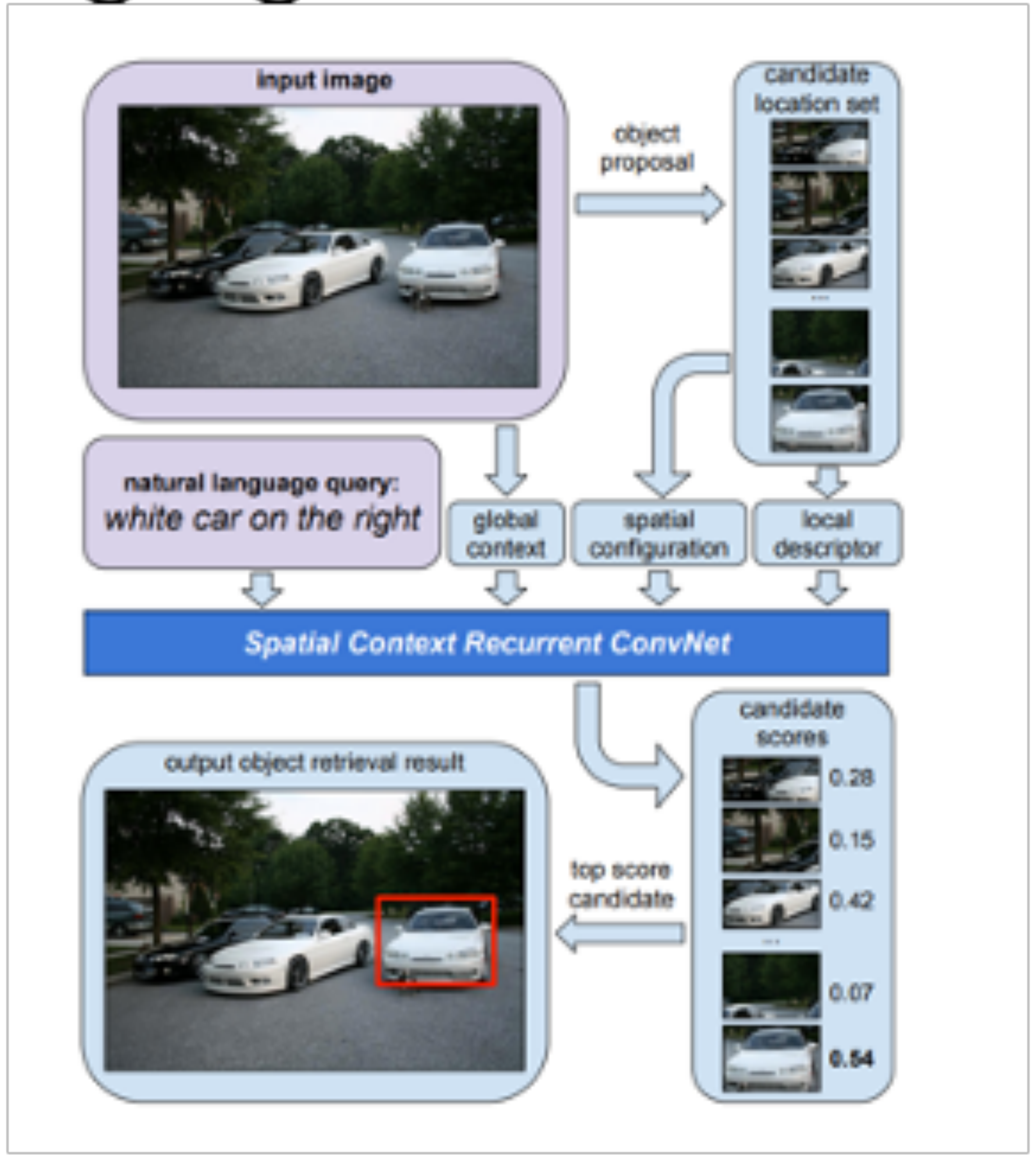
	Top-1 Acc (%)		AP@50 (%)	
Embedding	**DA-SJE**	**DS-SJE**	**DA-SJE**	**DS-SJE**
ATTRIBUTES	50.9	50.4	20.4	**50.0**
WORD2VEC	38.7	38.6	7.5	33.5
BAG-OF-WORDS	43.4	44.1	24.6	39.6
CHAR CNN	47.2	48.2	2.9	42.7
CHAR LSTM	22.6	21.6	11.6	22.3
CHAR CNN-RNN	54.0	54.0	6.9	45.6
WORD CNN	50.5	51.0	3.4	43.3
WORD LSTM	52.2	53.0	**36.8**	46.8
WORD CNN-RNN	**54.3**	**56.8**	4.8	48.7

CUB Birds

"Learning Deep Representations of Fine-Grained Visual Descriptions" (Reed et al, CVPR 2016)

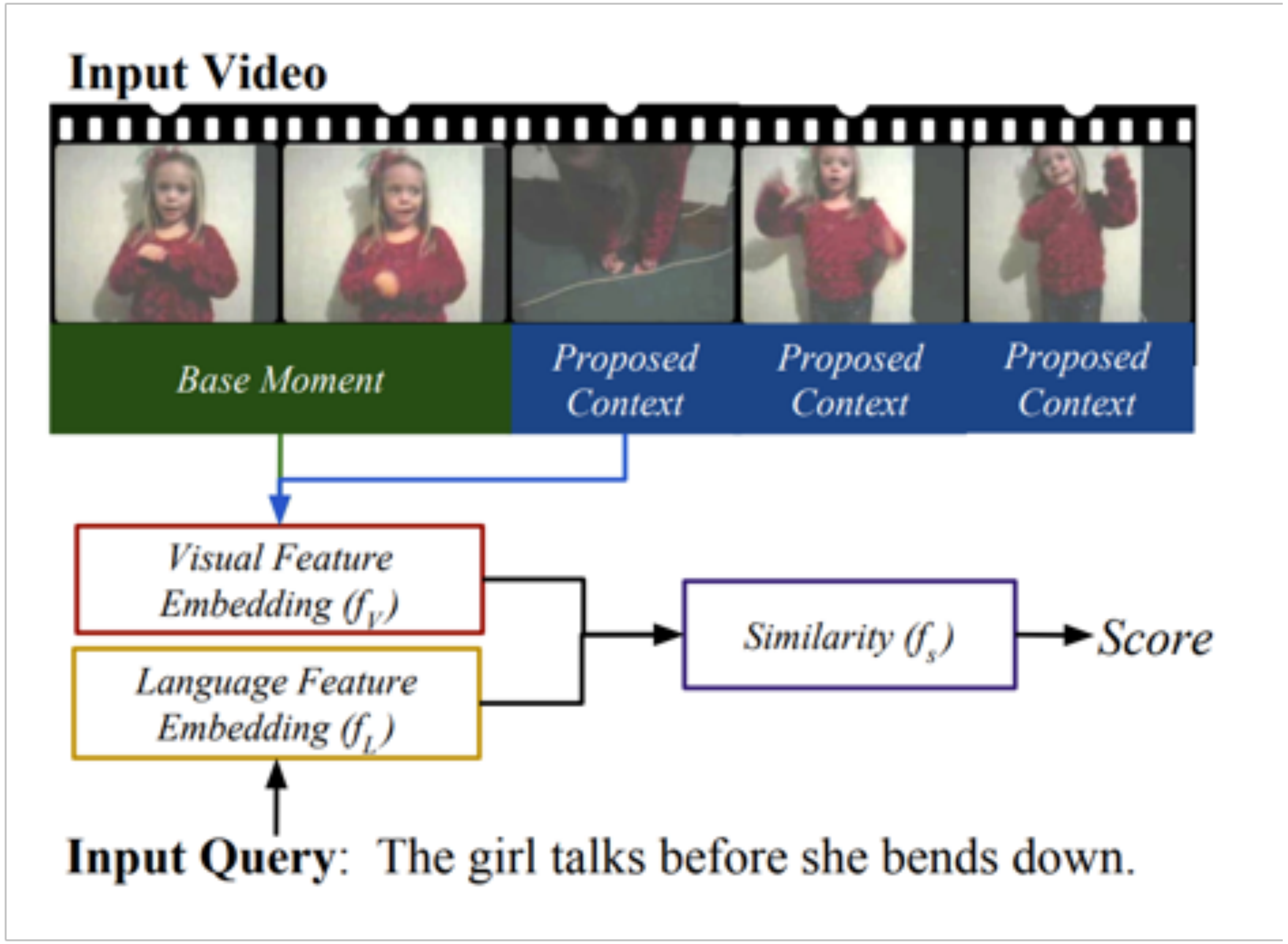
Grounding

Match image region to language



Natural Language Object Retrieval
(Hu et al, CVPR 2016)

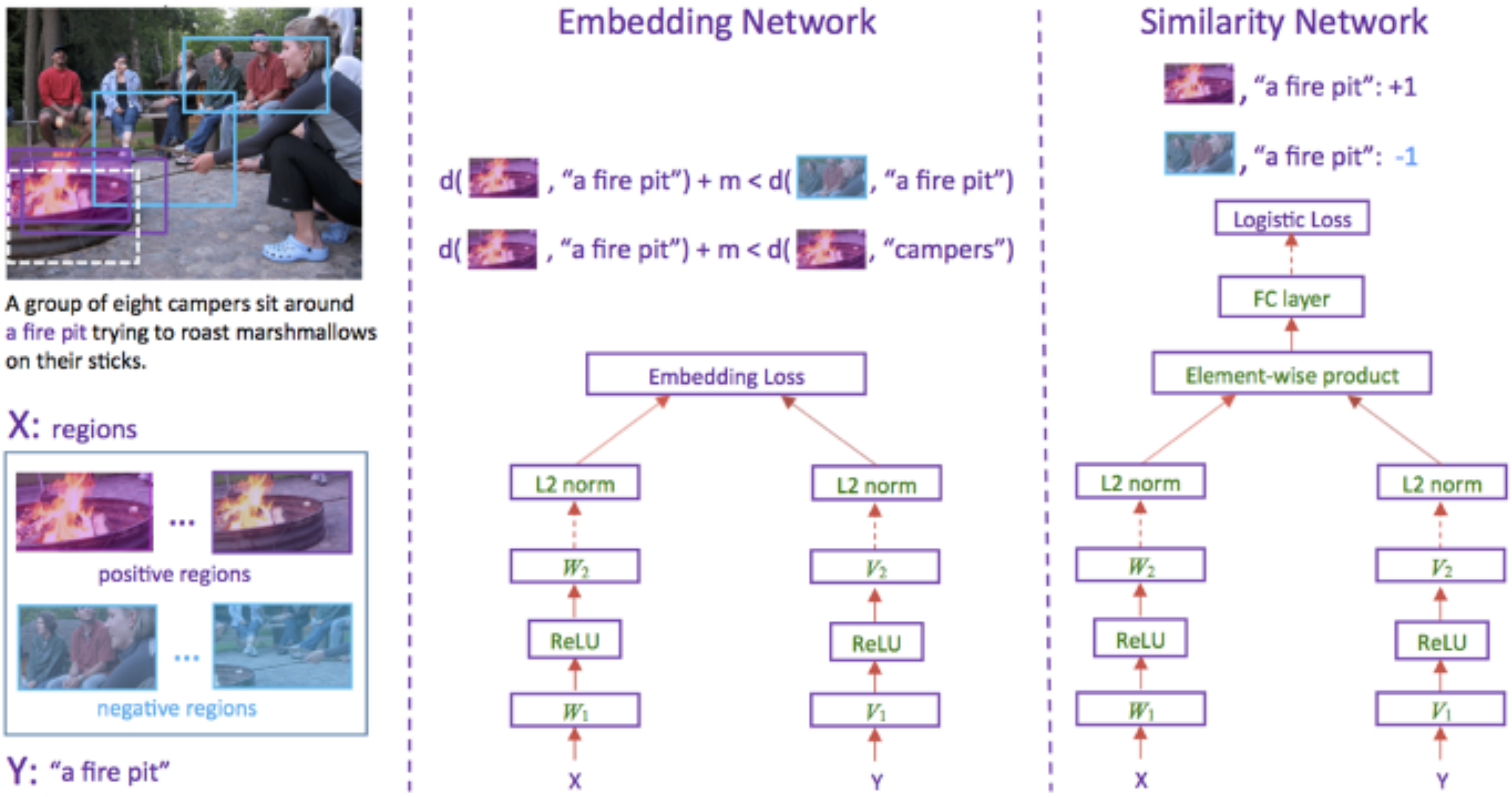
Match video frames to language



Localizing moments in video with temporal language
(Hendricks et al, EMNLP, 2018)

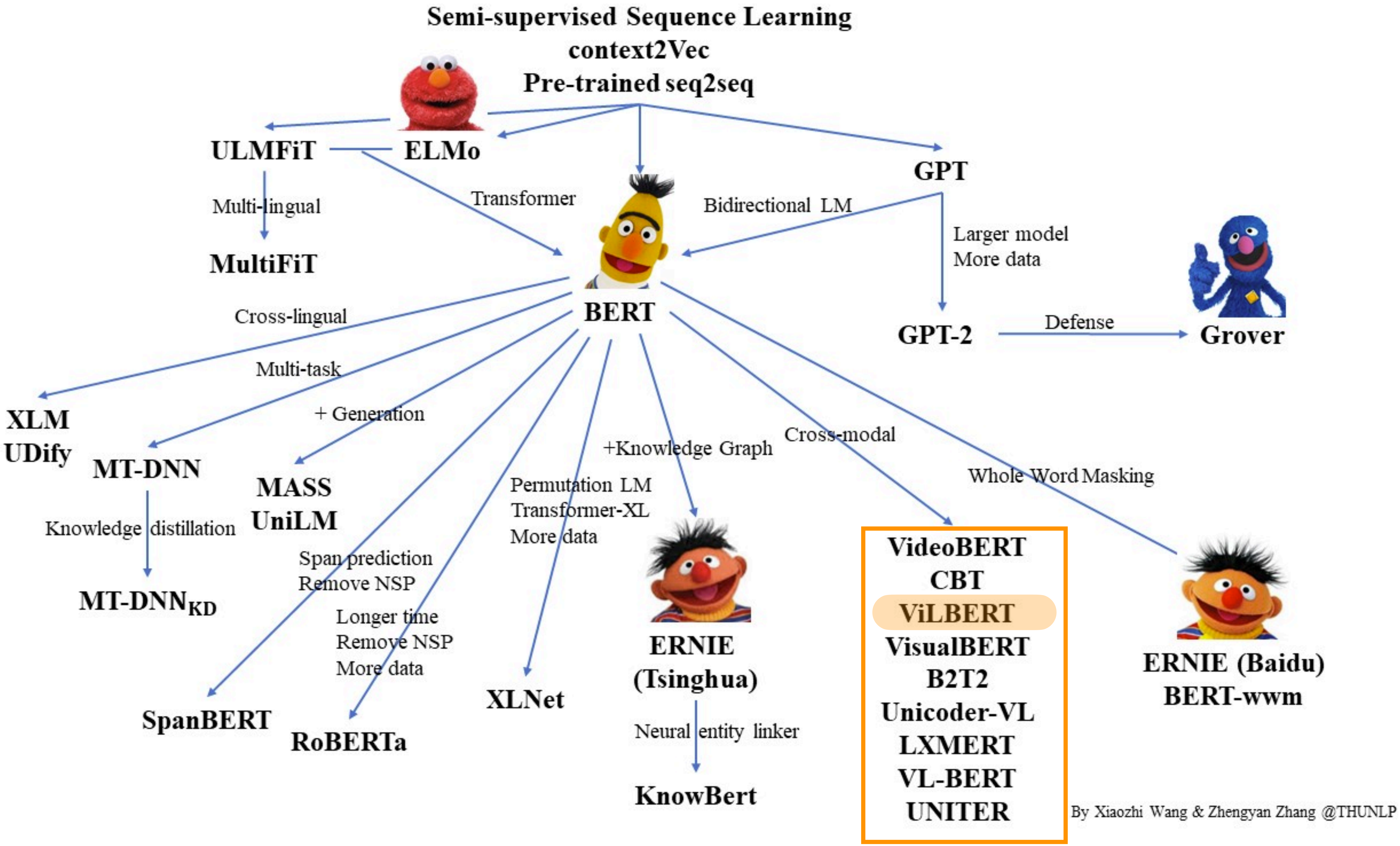
Grounding

Phrase Localization



Learning Two-Branch Neural Networks for Image-Text Matching Tasks

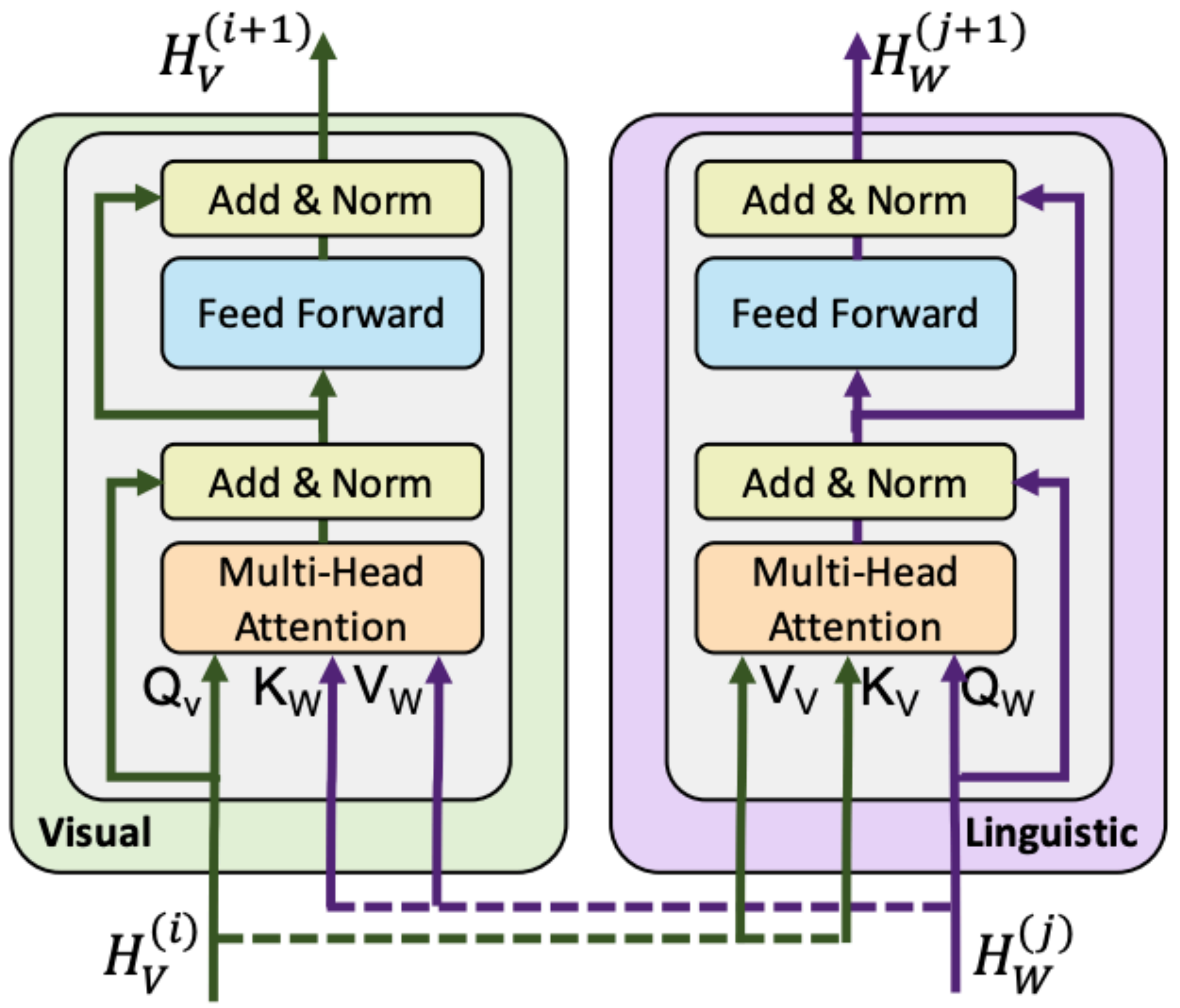
(Wang et al, TPAMI 2018)

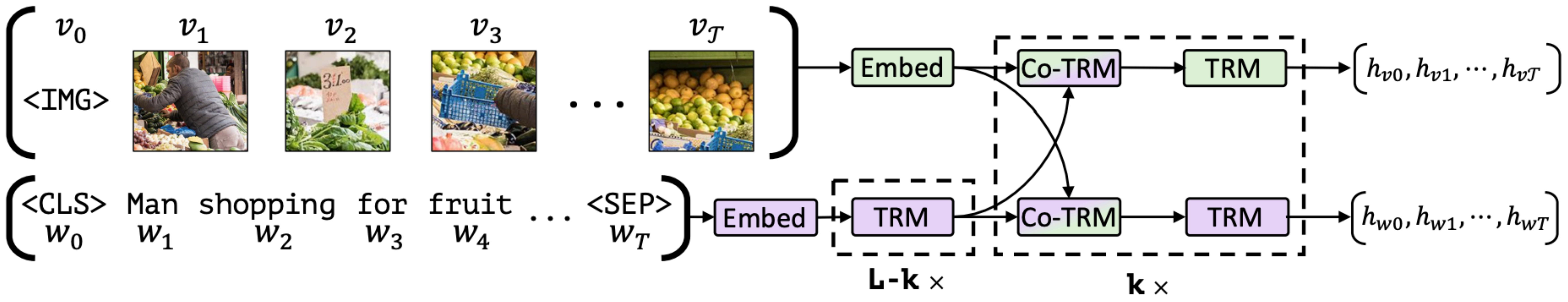

Semi-supervised Sequence Learning
context2Vec
Pre-trained seq2seq
ULMFiT
ELMo
GPT
Multi-lingual
Transformer
Bidirectional LM
Larger model
More data
MultiFiT
BERT
Cross-lingual
GPT-2
Defense
Grover
Multi-task
XLM
UDify
+ Generation
MT-DNN
MASS
UniLM
+Knowledge Graph
Cross-modal
Whole Word Masking
Permutation LM
Transformer-XL
More data
Knowledge distillation
Span prediction
Remove NSP
MT-DNN$_{KD}$
Longer time
Remove NSP
More data
ERNIE
(Tsinghua)
VideoBERT
CBT
ViLBERT
VisualBERT
B2T2
Unicoder-VL
LXMERT
VL-BERT
UNITER
ERNIE (Baidu)
BERT-wwm
SpanBERT
RoBERTa
XLNet
Neural entity linker
KnowBert
By Xiaozhi Wang & Zhengyan Zhang @THUNLP

Pretrained representations for vision and language

Image represented as

- series of **image region features** (extracted from pre-trained object detection network)
- **Region position** encoded as $5d$ vector





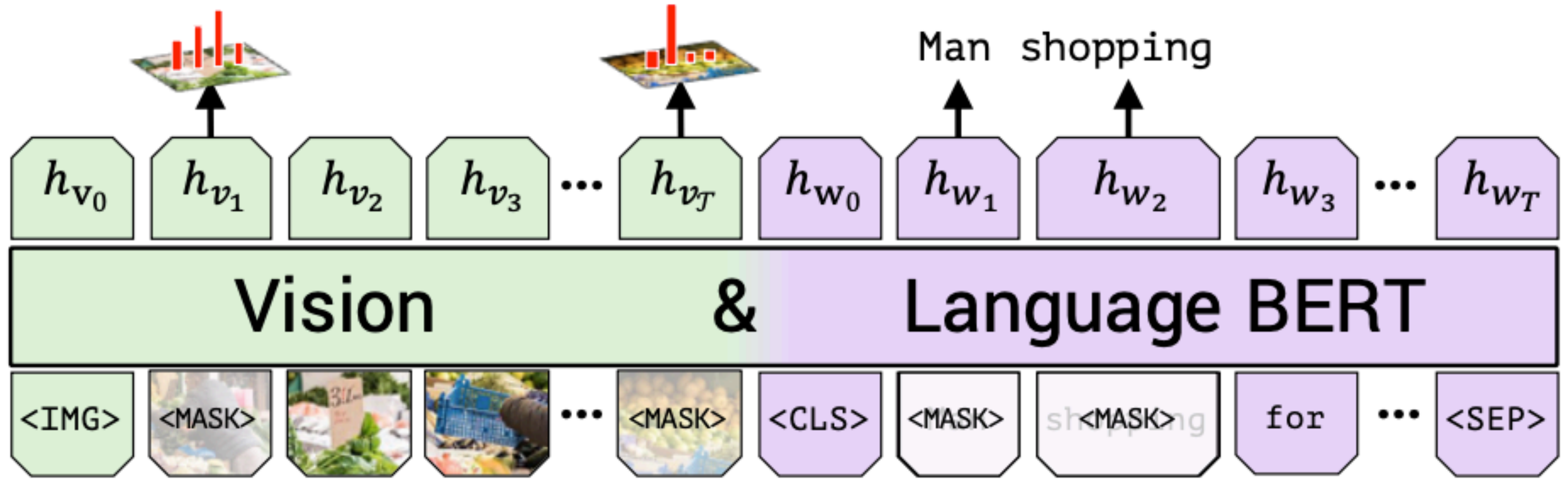
ViLBERT: Pretraining Task-Agnostic Visiolinguistic Representations for Vision-and-Language Tasks
[Lu et al 2019, https://arxiv.org/pdf/1908.02265.pdf]

Pretrained representations for vision and language

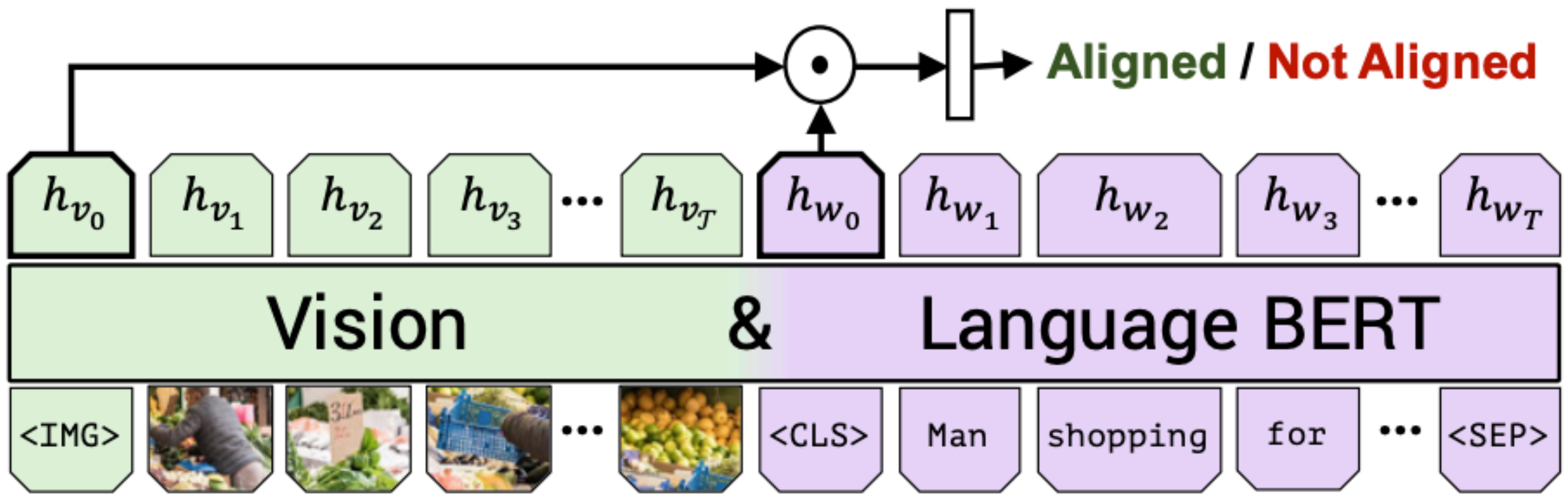
Predict semantic class distribution

Trained on

- Conceptual captions (~3.3M images with captions cleaned from alt-text labels)
- Two tasks to predict:
  - masked out words and semantic class distribution for masked out image regions
  - Is the image/description aligned?



(a) Masked multi-modal learning



(b) Multi-modal alignment prediction

ViLBERT: Pretraining Task-Agnostic Visiolinguistic Representations for Vision-and-Language Tasks
[Lu et al 2019, https://arxiv.org/pdf/1908.02265.pdf]

Pretrained representations for vision and language

		VQA [3]	VCR [25]			RefCOCO+ [32]			Image Retrieval [26]			ZS Image Retrieval		
	Method	test-dev (test-std)	Q→A	QA→R	Q→AR	val	testA	testB	R1	R5	R10	R1	R5	R10
SOTA	DFAF [36]	70.22 (70.34)	-	-	-	-	-	-	-	-	-	-	-	-
	R2C [25]	-	63.8 (65.1)	67.2 (67.3)	43.1 (44.0)	-	-	-	-	-	-	-	-	-
	MAttNet [33]	-	-	-	-	65.33	71.62	56.02	-	-	-	-	-	-
	SCAN [35]	-	-	-	-	-	-	-	48.60	77.70	85.20	-	-	-
Ours	Single-Stream†	65.90	68.15	68.89	47.27	65.64	72.02	56.04	-	-	-	-	-	-
	Single-Stream	68.85	71.09	73.93	52.73	69.21	75.32	61.02	-	-	-	-	-	-
	ViLBERT†	68.93	69.26	71.01	49.48	68.61	75.97	58.44	45.50	76.78	85.02	0.00	0.00	0.00
	ViLBERT	**70.55 (70.92)**	**72.42 (73.3)**	**74.47 (74.6)**	**54.04 (54.8)**	**72.34**	**78.52**	**62.61**	**58.20**	**84.90**	**91.52**	**31.86**	**61.12**	**72.80**

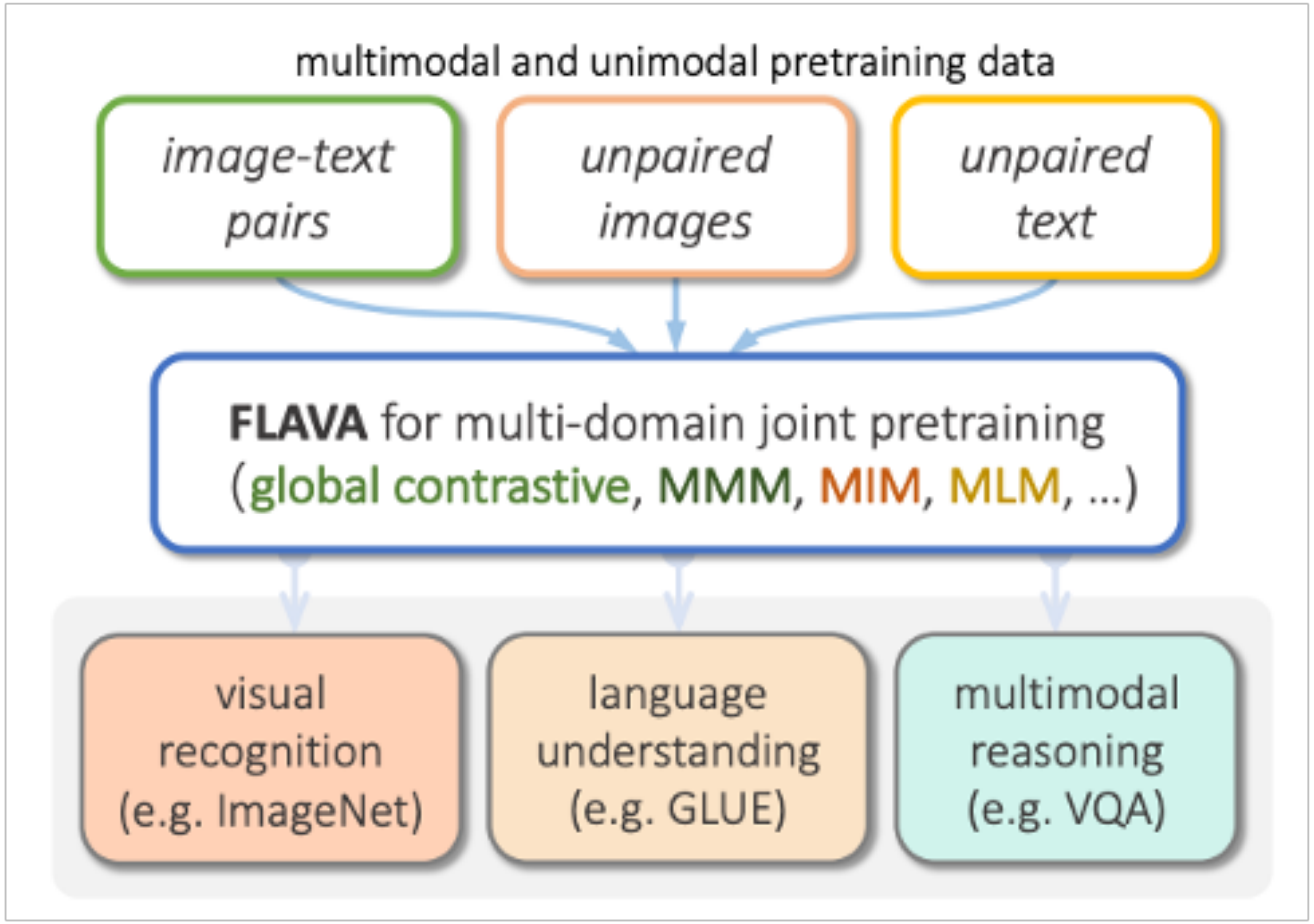
Pretraining improves performance on variety of vision+language tasks!


ViLBERT

ViLBERT: Pretraining Task-Agnostic Visiolinguistic Representations for Vision-and-Language Tasks
[Lu et al 2019, https://arxiv.org/pdf/1908.02265.pdf]

Large multi-modal models

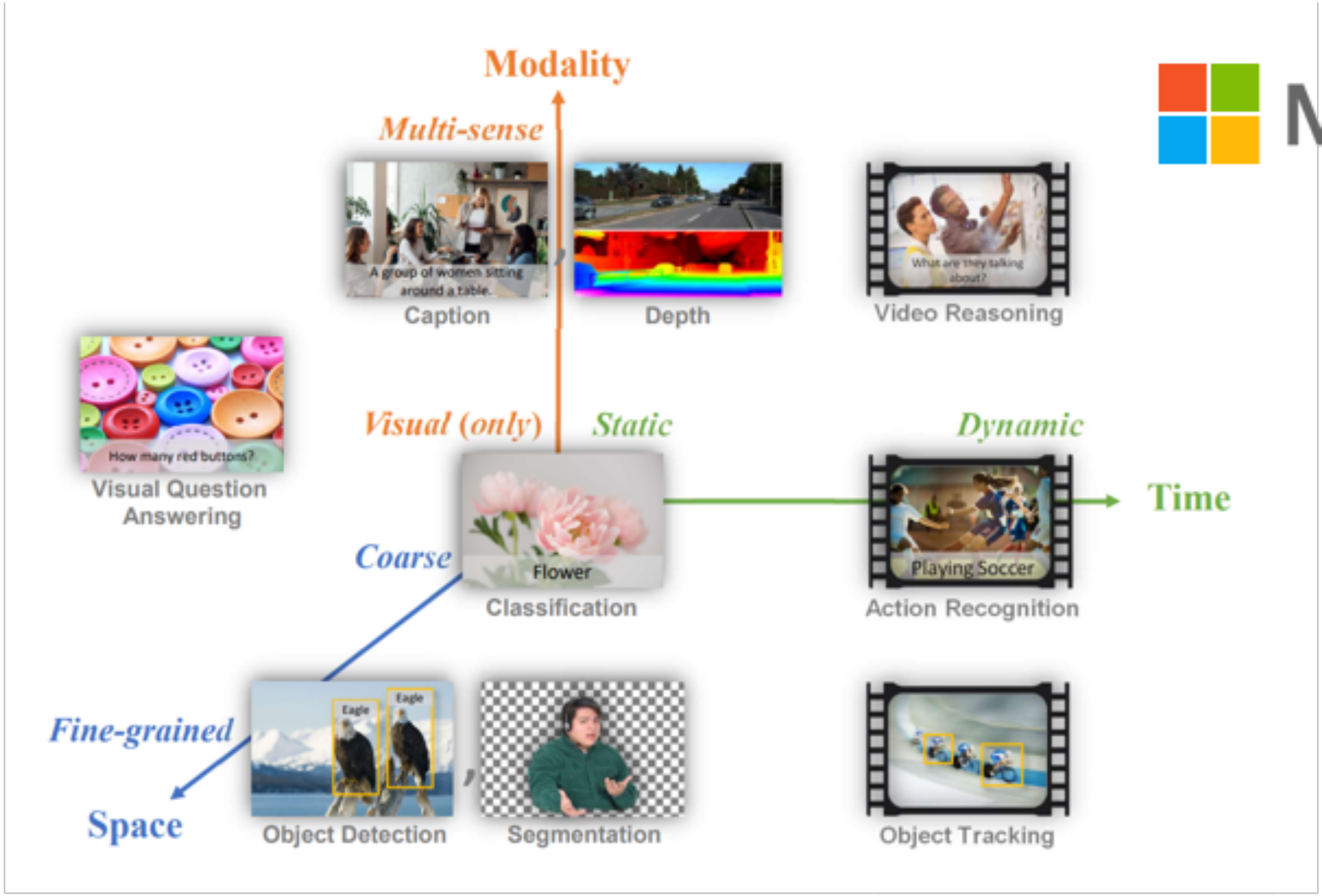
FLAVA



FLAVA: A Foundational Language And Vision Alignment Model (Singh et al, CVPR 2022)

Large multi-modal, multi-lingual models

Florence



Microsoft

Florence: A New Foundation Model for Computer Vision (Yuan et al, CVPR 2022)

Some grounding tasks

- **Vision**
  - Captioning
  - Text to image generation and manipulation
  - Visual question answering (VQA)
  - Referring Expressions and Spatial reasoning
- **Interaction**
  - Instruction following
  - Text-based games

Image captioning

the girl is licking the spoon of batter

- Describe an image in a sentence

Image captioning

the girl is licking the spoon of batter

- Describe an image in a sentence
- Requires recognizing objects, attributes, relations in image
- Caption must be fluent

(MS COCO, Chen et al., 2015)

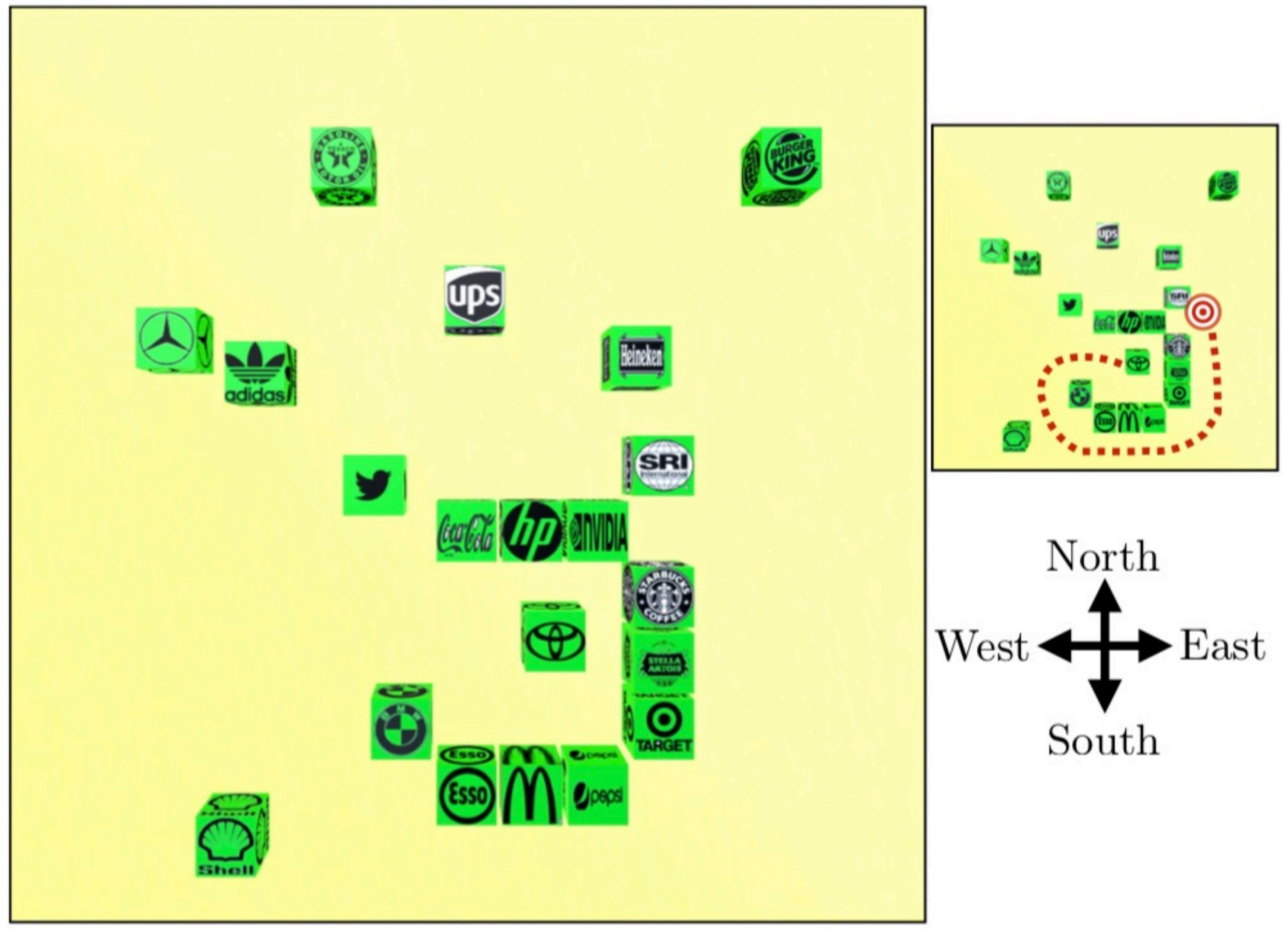
Captioning as multi-modal translation

French → Text encoder (e.g. RNN) → Text decoder (e.g. RNN) → *English*

 → Image encoder → Text decoder (e.g. RNN) → *English*

(Donahue et al., 2015, Vinyals et al., 2015)

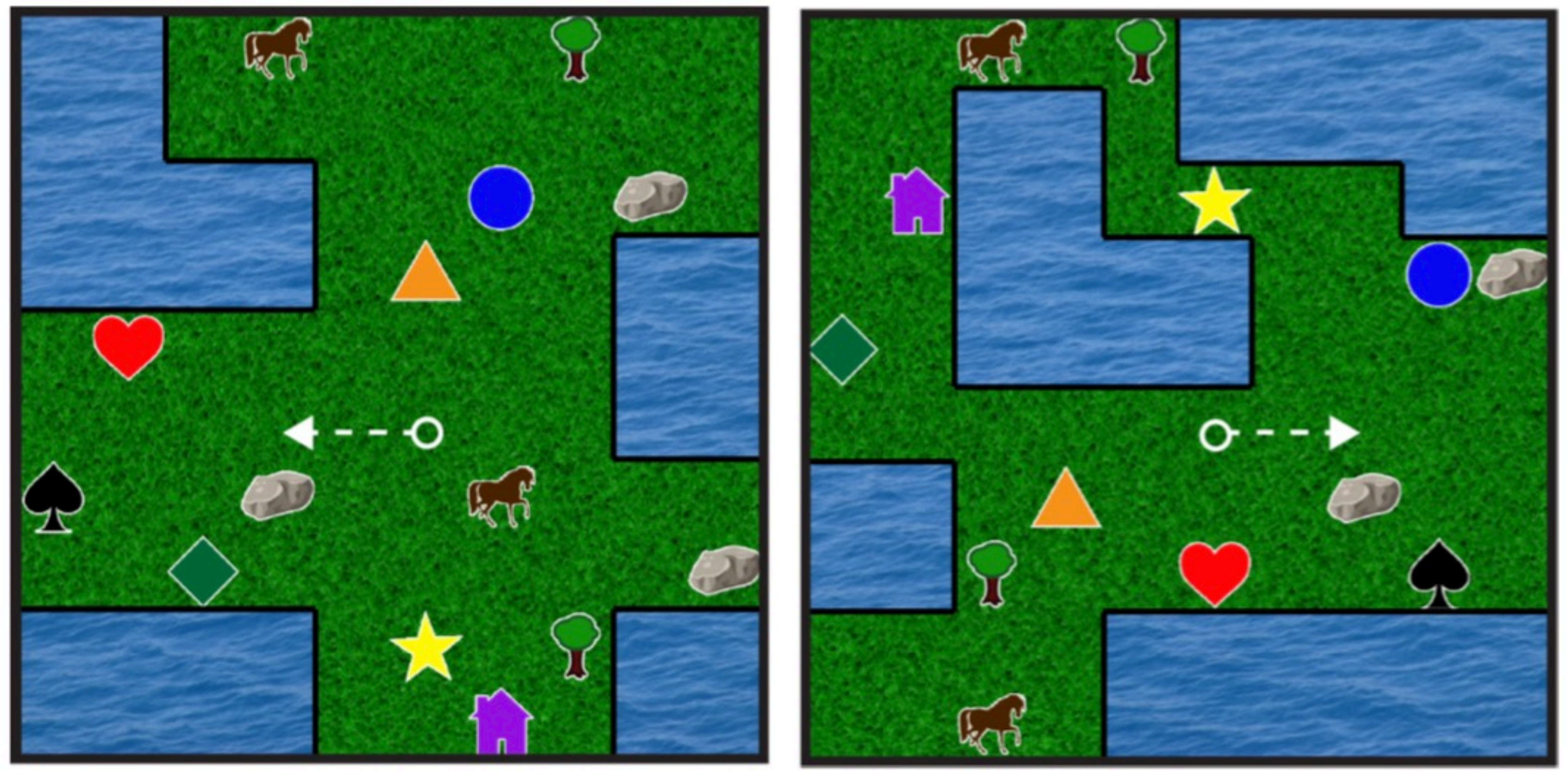
Spatial Reasoning



Put the Toyota block in the same row as the SRI block, in the first open space to the right of the SRI block
Move Toyota to the immediate right of SRI, evenly aligned and slightly separated
Move the Toyota block around the pile and place it just to the right of the SRI block
Place Toyota block just to the right of The SRI Block
Toyota, right side of SRI

Robotic Manipulation

(Bisk et al., 2016, Misra et al., 2017)

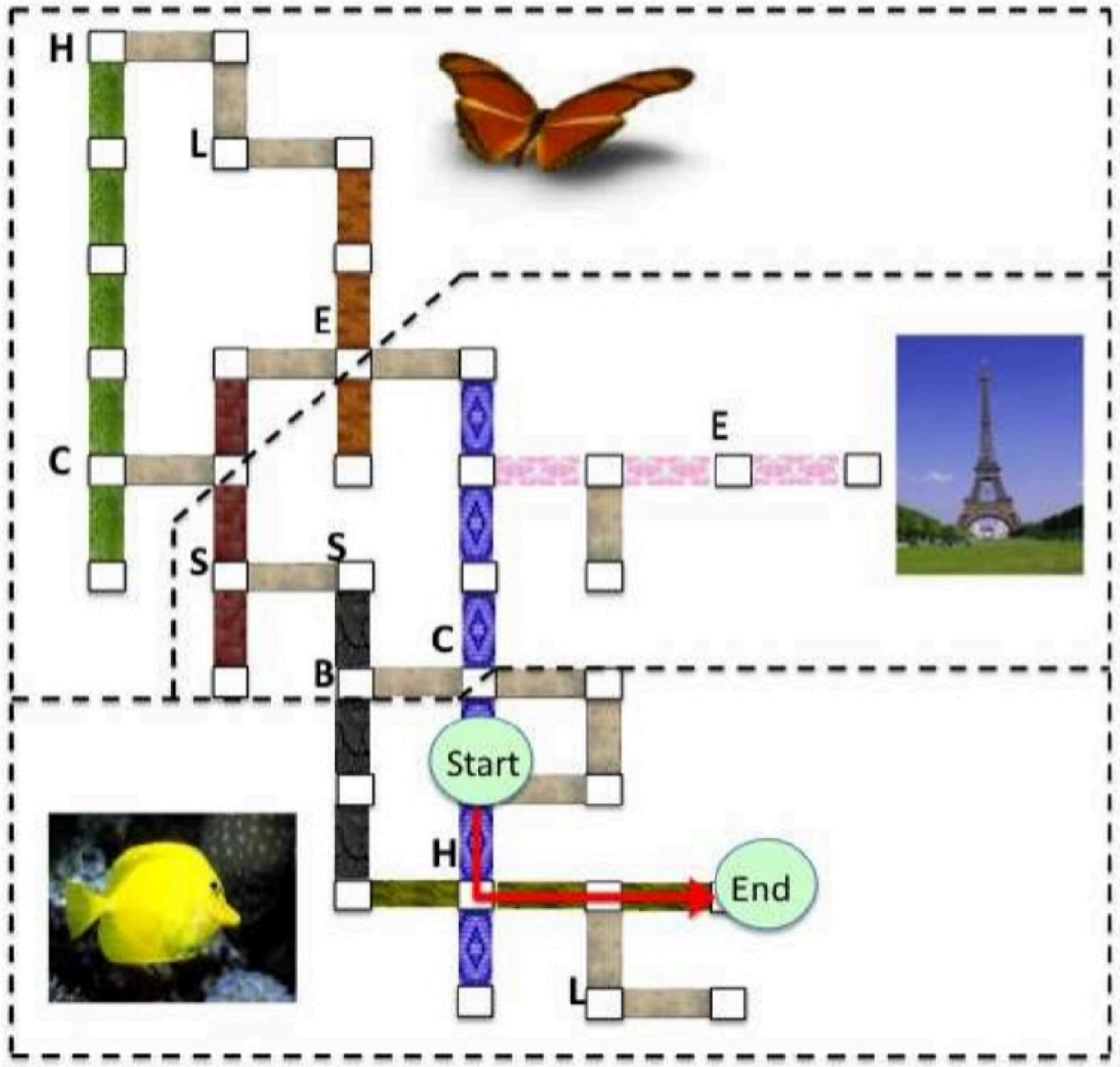


Reach the cell above the westernmost rock

Autonomous navigation

(Janner et al., 2017)

Instruction Following

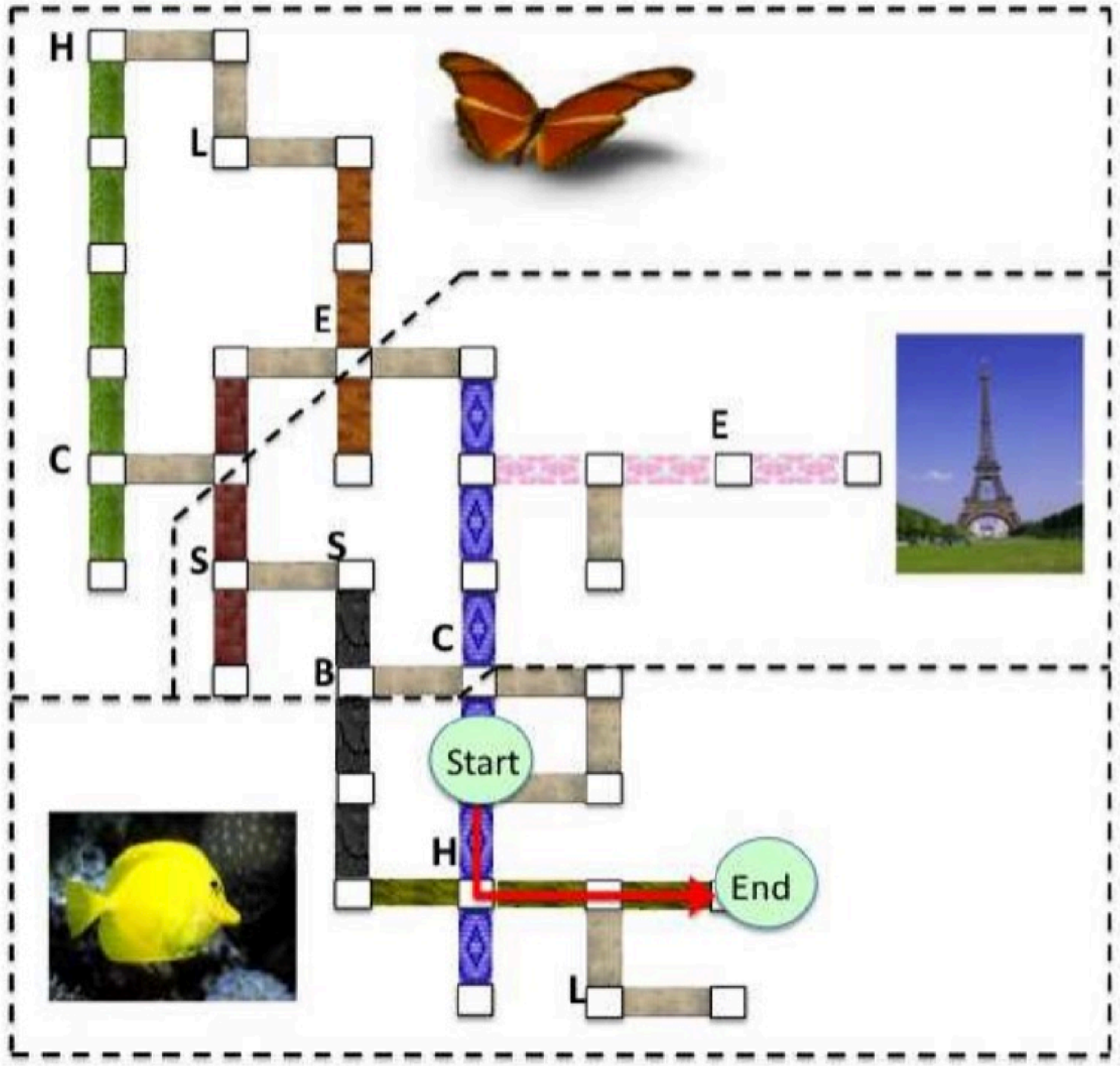


- Want to be able to follow instructions in a virtual environment
- "Go along the blue hall, then turn left away from the fish painting and walk to the end of the hallway"

(MacMahon et al., 2006)

(slide adapted from Greg Durrett)

Instruction Following



Instruction: "Go away from the lamp to the intersection of the red brick and wood"

Basic: Turn ( ) ,
Travel (steps: 1)

Landmarks: Turn ( ) ,
Verify (left: WALL , back: LAMP , back: HATRACK , front: BRICK HALL) ,
Travel (steps: 1) ,
Verify (side: WOOD HALL)

- Train semantic parser on (utterance, action) pairs
- Language is grounded in actions in the world

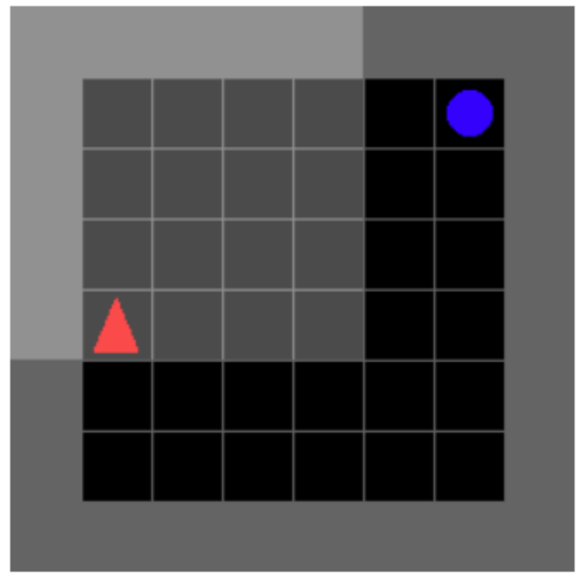
(Chen and Mooney, 2011)

(slide adapted from Greg Durrett)

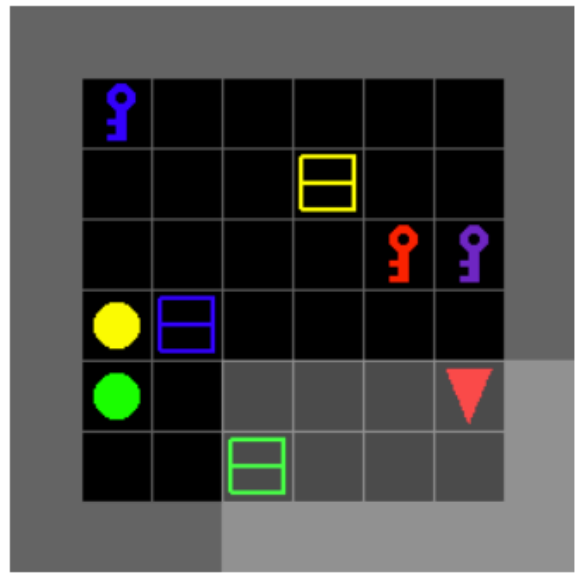
Frameworks for understanding grounded language (with perception and actions)

BabyAI

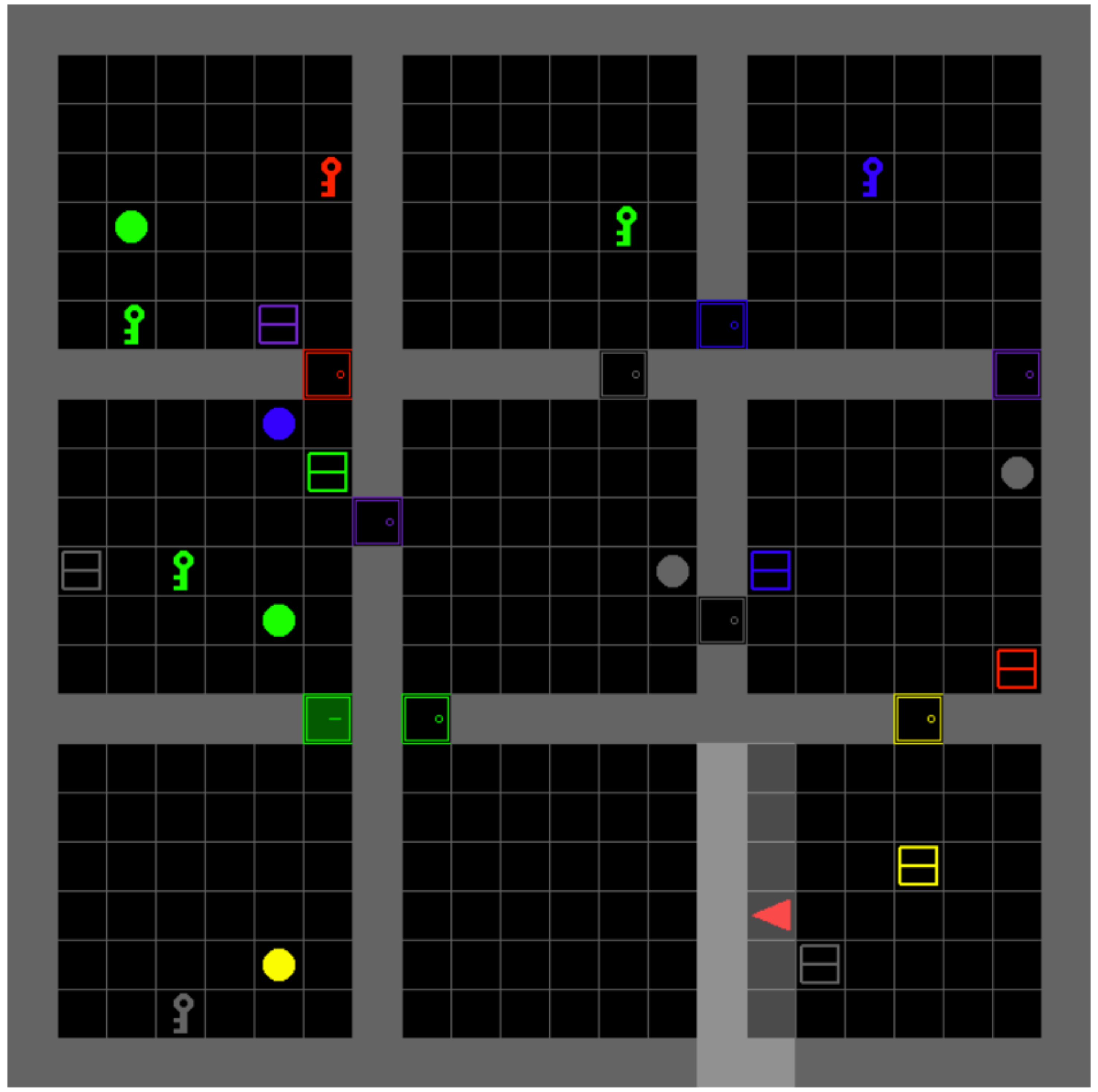
- Grid Environment
- Generated (synthetic language) using grammar
- Easy to hard levels
- Studies grounding and compositionality



(a) GoToObj: "go to the blue ball"



(b) PutNextLocal: "put the blue key next to the green ball"

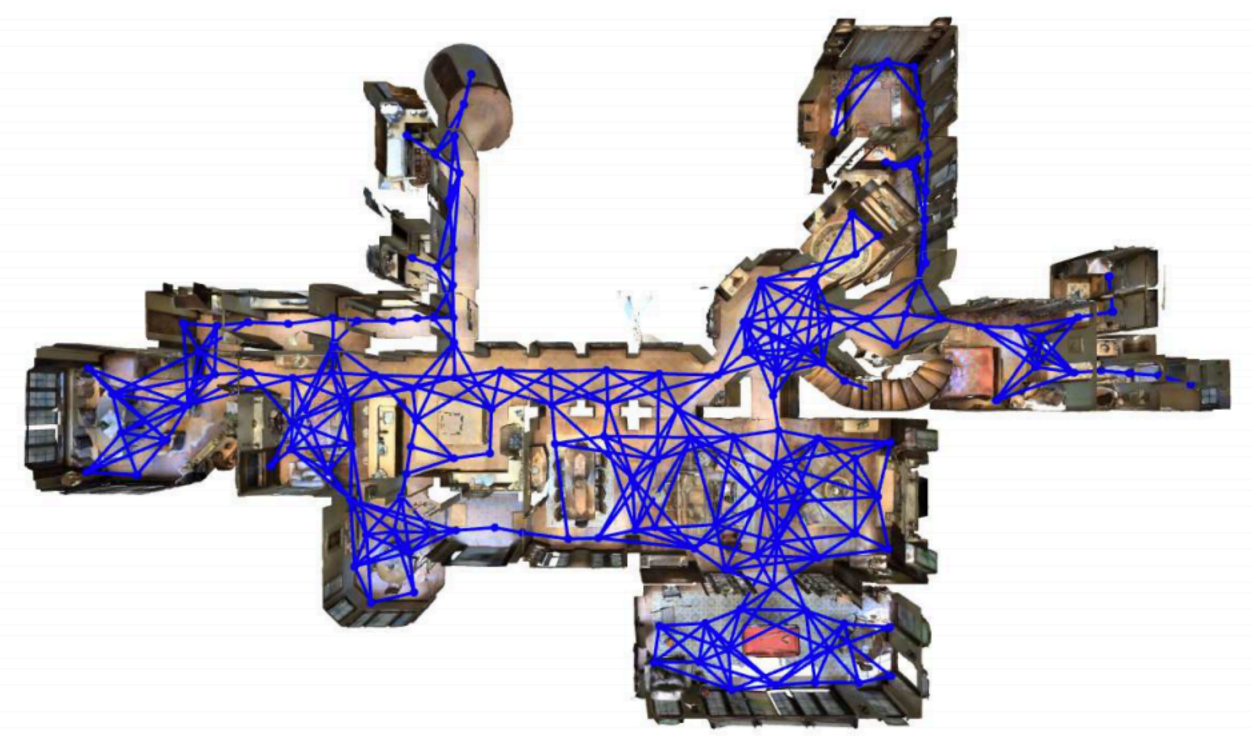


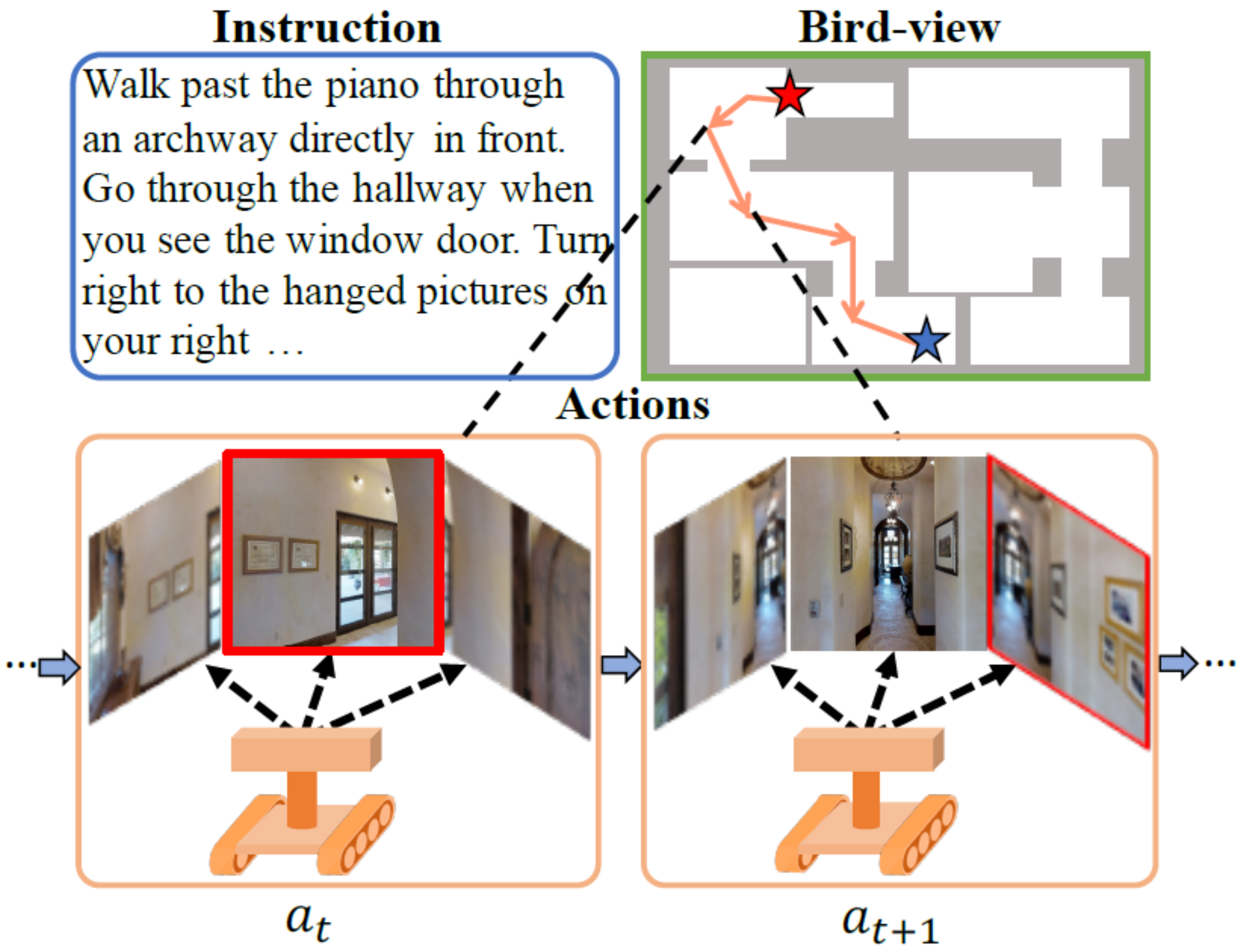
(c) BossLevel: "pick up the grey box behind you, then go to the grey key and open a door". Note that the green door near the bottom left needs to be unlocked with a green key, but this is not explicitly stated in the instruction.

BabyAI: A Platform to Study the Sample Efficiency of Grounded Language Learning
[Chevalier-Boisvert et al 2018, https://arxiv.org/pdf/1810.08272.pdf]

Vision-and-language Navigation

- More realistic houses
- Human instructions navigation
- Discrete action space
- Navigation graph

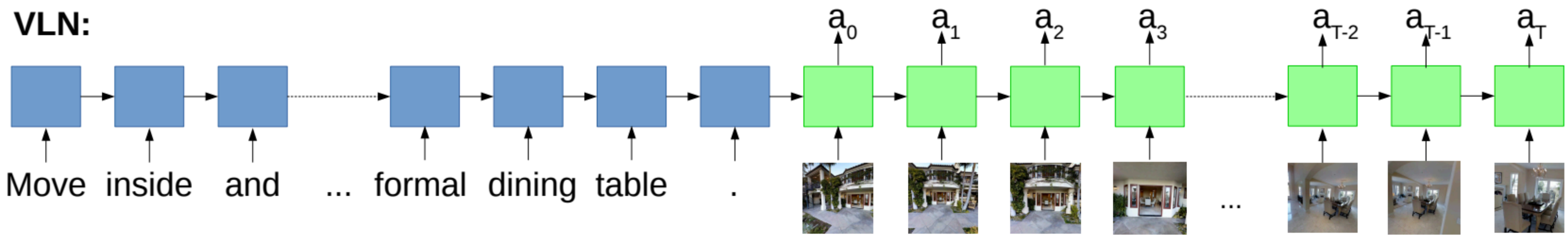




Vision-and-Language Navigation: Interpreting visually-grounded navigation instructions in real environments [Anderson et al 2018, https://bringmeaspoon.org/]

Vision-and-language Navigation

- Sequence of words to sequence of actions!



Input Images at each time step

Vision-and-Language Navigation: Interpreting visually-grounded navigation instructions in real environments
[Anderson et al 2018, https://bringmeaspoon.org/]

Vision-and-language Navigation

Leave the bedroom, and enter the kitchen. Walk forward, and take a left at the couch. Stop in front of the window.

ALFRED

- More realistic houses
- Sequence of human instructions for common household tasks
- Study embodied language understanding

ALFRED: A Benchmark for Interpreting Grounded Instructions for Everyday Tasks
[Shridhar et al 2019, https://askforalfred.com/]

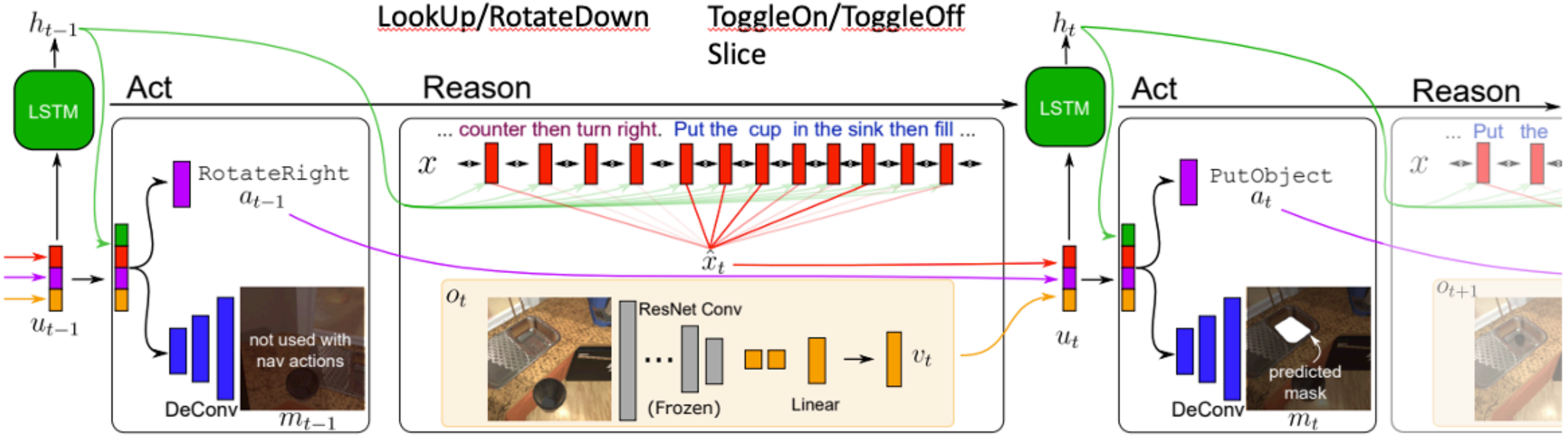
ALFRED agent model

- Seq2seq model (CNN vision, LSTM language)
- Predicts action + binary mask of object from concatenated input
  - 13 actions (5 navigation + 7 interaction + stop)

Concatenated input
Vision
Language
Last action

Navigation
MoveAhead
RotateLeft/RotateRight
LookUp/RotateDown

Interaction
Pickup/Put
Open/Close
ToggleOn/ToggleOff
Slice



CMPT 983

Grounded Natural Language Understanding